Axiomatic Models of Constitutional Interpretation: Originalism, Living Constitutionalism, Textualism, Structuralism, and Pragmatism

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Abstract

I formalize five leading theories of constitutional interpretation—originalism, living constitutionalism, textualism, structuralism, and pragmatism—as axiomatic decision models sharing a common set of primitives. Each theory is characterized by a distinctive objective function and constraint set, reflecting its core commitments regarding the source of legal authority, the treatment of textual ambiguity, and the permissible role of subjective judicial values. By translating these interpretive frameworks into comparable mathematical structures, I clarify the extent to which each can limit judicial discretion. The models reveal that while originalism and textualism impose the tightest formal constraints, no theory eliminates discretion entirely; textual ambiguity and the inescapable need for normative judgment ensure that residual discretion persists in every constitutional decision. I prove an impossibility theorem showing that complete elimination of discretion is unattainable under any axiomatic interpretation model. I further demonstrate that values and ambiguity interact multiplicatively, with ambiguity amplifying the influence of values. I conclude that the choice among interpretive theories is itself an irreducibly normative act that cannot be resolved by formal constraint alone.

Keywords: Constitutional interpretation, originalism, living constitutionalism, textualism, structuralism, pragmatism, judicial discretion, axiomatic models

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1 Introduction

Constitutional interpretation in the United States is shaped by a family of competing theories, each advancing a distinct account of where legal authority resides and how Justices should resolve the pervasive ambiguities of the constitutional text. In this article, I provide a rigorous examination of the five principal theories of constitutional interpretation in American law—originalism, living constitutionalism, textualism, structuralism, and pragmatism. My central claim is that despite decades of rich doctrinal and philosophical debate, the field lacks a formal apparatus that renders these theories commensurable and exposes their precise implications for judicial discretion. I supply such an apparatus.

The principal contenders differ not only in their historical pedigree and intellectual champions but also in their implicit assumptions about the nature of law, the role of the judiciary, and the legitimacy of value-laden judgment. For each theory, I define its core principles, trace its historical evolution through seminal scholarly works, and quote key passages to capture the theory's animating commitments. I then identify major Supreme Court decisions that explicitly or implicitly rely on each interpretive method, detailing the opinion author, holding, and principal dissenting views. By mapping the interplay between theoretical articulation and judicial practice, the analysis illuminates how competing accounts of legal authority shape the development of constitutional doctrine. I conclude that while each theory offers a distinct vision of the judicial role, none fully escapes the influence of normative judgment, and the persistent debate among them reflects deep, unresolved tensions about the nature of constitutional legitimacy.

I construct a unified axiomatic framework in which each interpretive theory is represented as a constrained optimization problem faced by a Justice. The model setup is held constant across theories: a Justice J confronts a constitutional provision T that exhibits textual ambiguity A, operates within a context C (historical, structural, precedential), and brings to the task a set of subjective values V. The Justice selects a decision d from a feasible set $\mathcal{D}$. The differences among theories are captured by the objective function that d must maximize and the constraints that define the admissible decision set. This formalization allows me to examine, with precision, the role of subjective values and the capacity of each theory to cabin judicial discretion. I prove that complete elimination of judicial discretion is impossible under any axiomatic model, and that textual ambiguity systematically amplifies the influence of subjective values.

The paper is organized as follows. Section 2 recapitulates the core principles, historical evolution, and major defenders of the five theories. Section 3 develops the common axiomatic framework and specializes it for each theory, presenting the key equations and proving two formal results: a constraint hierarchy and a value-penetration ordering. Section 4 analyzes the interaction of subjective values and textual ambiguity within each model, formalizing value influence functions, the ambiguity vector, and proving that ambiguity amplifies value influence. Section 5 assesses whether judicial discretion can be limited under any of the formalized theories, defines discretion measures, proves an impossibility theorem, and addresses the meta-choice problem. Section 6 concludes. All proposition, theorem, and corollary proofs are collected in the Annex.

2 The Five Frameworks of Constitutional Interpretation

American constitutional law is shaped by a small set of competing interpretive theories, each advancing a distinct account of where legal authority resides and how Justices should resolve the pervasive ambiguities of the constitutional text. Five frameworks dominate the scholarly and judicial conversation: originalism, living constitutionalism, textualism, structuralism, and pragmatism. These theories differ not only in their methodological prescriptions but also in their underlying assumptions about the nature of law, the role of the judiciary, and the legitimacy of value-laden judgment. This article offers a detailed exposition of each theory's core principles, traces its historical evolution through its most influential scholarly statements, and demonstrates its operation in landmark Supreme Court decisions. By juxtaposing theoretical commitments with judicial application, the article reveals both the internal logic and the practical tensions of each interpretive mode.

2.1 Originalism

Core Principles

Originalism holds that the meaning of the Constitution is fixed at the time of its ratification and that this fixed meaning binds subsequent interpreters. Early formulations emphasized the original intentions of the Framers—what the drafters subjectively meant to accomplish. Contemporary "new originalism" focuses on the original public meaning: the understanding that a reasonable speaker of English at the time of enactment would have ascribed to the words of the text (Solum 2008). The core implication is that constitutional change must occur through the Article V amendment process, not through judicial updating. Originalism thus functions as a theory of constraint: it seeks to tether judicial discretion to a historically determinable semantic baseline.

Historical Evolution and Seminal Works

Originalism emerged as a self-conscious movement in the 1970s and 1980s, largely in reaction to the expansive rights jurisprudence of the Warren and early Burger Courts. Robert H. Bork's "The Tempting of America" (1990) provided a politically charged defense of original intent, arguing that the Constitution's legitimacy depends on its ratification by the people and that Justices must enforce only the value choices made at that founding moment. Bork wrote: "The only legitimate basis for constitutional adjudication is the original intent of the framers and ratifiers. Any other basis allows the Justice to impose his own values on the society" (Bork 1990, 143).

Justice Antonin Scalia became the theory's most prominent judicial advocate, shifting the focus from subjective intent to objective textual meaning. In "A Matter of Interpretation" (1997), Scalia declared: "The Constitution that I interpret and apply is not living but dead—or, as I prefer to call it, enduring. It means today not what current society, much less the Court, thinks it ought to mean, but what it meant when it was adopted" (Scalia 1997, 38). This commitment to original public meaning was further developed by scholars such as Randy Barnett, who in "Restoring the Lost Constitution" (2004) argued that the original meaning of the Constitution establishes a "presumption of liberty" that constrains governmental power. Barnett asserted: "The original meaning of the Constitution provides the law that governs those who govern us" (Barnett 2004, 4). Lawrence B. Solum's work on "semantic originalism" (2008) provided a sophisticated linguistic framework, distinguishing between semantic meaning and expected applications, thereby refining the theory's response to charges of rigidity.

Major Supreme Court Decision: District of Columbia v. Heller (2008)

The Supreme Court's decision in District of Columbia v. Heller, 554 U.S. 570 (2008), stands as the most prominent originalist ruling in modern times. Justice Scalia wrote the opinion of the Court, holding that the Second Amendment protects an individual right to keep and bear arms for self-defense, unconnected with militia service. The opinion canvassed the text, founding-era dictionaries, state constitutions, and post-ratification commentary to establish the original public meaning of the amendment. Scalia concluded: “The Second Amendment’s text, as informed by its historical background, compels the conclusion that the citizen’s right to keep and bear arms is an individual right” (Heller, 554 U.S. at 595).

Justice Stevens dissented, joined by Justices Souter, Ginsburg, and Breyer. Stevens argued that the original understanding of the Second Amendment protected only a militia-related right and that the majority’s historical analysis was selective. Justice Breyer filed a separate dissent, joined by the same three Justices, advocating an interest-balancing approach that would have upheld the District’s handgun ban as a reasonable regulation. The Heller decision exemplifies originalism in action, but the sharp disagreement among the Justices over the historical record also illustrates the epistemic challenges that the theory faces when the evidence of original meaning is contested.

2.2 Living Constitutionalism

Core Principles

Living constitutionalism maintains that the Constitution’s meaning evolves over time to reflect changing societal values, circumstances, and moral insights. It rejects the notion of a fixed semantic anchor, treating the text as a broad charter of principles whose content is filled in by each generation. On this view, the Constitution endures precisely because it is not tied to the specific expectations of the eighteenth century; its abstract phrases—such as “due process,” “equal protection,” and “cruel and unusual punishment”—invite Justices to apply enduring principles to contemporary problems. The theory draws on the common-law tradition, legal realism, and the philosophical claim that law is an interpretive practice aimed at moral justification.

Historical Evolution and Seminal Works

The intellectual roots of living constitutionalism lie in the Progressive Era’s critique of formalism and in the legal realist insight that Justices inevitably make law. Justice William J. Brennan gave the theory its most famous judicial articulation in a 1985 speech, later published as “The Constitution of the United States: Contemporary Ratification.” Brennan argued: “The genius of the Constitution rests not in any static meaning it might have had in a world that is dead and gone, but in the adaptability of its great principles to cope with current problems and current needs” (Brennan 1985, 438).

David A. Strauss’s "The Living Constitution" (2010) provided a systematic defense grounded in the common-law method. Strauss contended that “the living Constitution is a common law Constitution. It evolves over time through a process of deciding cases, in which Justices build on what earlier Justices have done” (Strauss 2010, 3). This incremental, precedent-based evolution, Strauss argued, produces more stable and democratically responsive outcomes than originalism. Ronald Dworkin’s “moral reading” of the Constitution, articulated in "Freedom’s Law" (1996), supplied a philosophical foundation. Dworkin proposed that abstract constitutional clauses refer to moral concepts, and Justices must decide cases by constructing the best moral theory that fits and justifies the constitutional text and past decisions. He wrote: “The moral reading proposes that we all—Justices, lawyers, citizens—interpret and apply these abstract clauses on the understanding that they invoke moral principles about political decency and justice" (Dworkin 1996, 2).

Major Supreme Court Decision: Obergefell v. Hodges (2015)

The Supreme Court's decision in Obergefell v. Hodges, 576 U.S. 644 (2015), is a paradigmatic example of living constitutionalism. Justice Kennedy wrote the opinion of the Court, holding that the Fourteenth Amendment requires states to license and recognize same-sex marriages. Kennedy's opinion traced the evolution of the Court's liberty jurisprudence and emphasized that the understanding of marriage had changed over time. He wrote: "The nature of injustice is that we may not always see it in our own times. The generations that wrote and ratified the Bill of Rights and the Fourteenth Amendment did not presume to know the extent of freedom in all of its dimensions, and so they entrusted to future generations a charter protecting the right of all persons to enjoy liberty as we learn its meaning" (Obergefell, 576 U.S. at 664). This passage encapsulates the living constitutionalist commitment to an evolving understanding of fundamental rights.

Chief Justice Roberts dissented, joined by Justices Scalia and Thomas. Roberts argued that the Court had substituted its own policy preferences for the democratic process, and that the Constitution does not address the definition of marriage. Justice Scalia filed a separate dissent, joined by Justice Thomas, condemning the majority's reasoning as a "threat to American democracy" and a usurpation of legislative power. Justice Thomas, joined by Justice Scalia, argued that the Due Process Clause protects only freedom from physical restraint and does not confer substantive rights. Justice Alito, joined by Justices Scalia and Thomas, emphasized that the traditional definition of marriage was deeply embedded in American law and culture. The case illustrates both the transformative potential of living constitutionalism and the fierce opposition it provokes from those who insist on a more constrained judicial role.

2.3 Textualism

Core Principles

Textualism insists that interpretation must be grounded in the ordinary meaning of the constitutional text, read in its entirety and with attention to linguistic context, structure, and established canons of construction. Although often allied with originalism, textualism is a distinct modality that prioritizes the semantic content of the enacted words over external historical evidence of intent or purpose. Textualists argue that the written text is the only law that was democratically enacted, and Justices must enforce it as written, not as they wish it had been written. The methodology relies on dictionaries, grammar, usage conventions, and canons such as *noscitur a sociis* and *ejusdem generis* to resolve ambiguity, and it generally excludes legislative history as an authoritative source of meaning.

Historical Evolution and Seminal Works

Modern textualism emerged in the 1980s as a reaction against purposivist and legislative-history approaches that had dominated statutory interpretation. Justice Scalia's "A Matter of Interpretation" (1997) is the theory's most influential statement. Scalia wrote: "The text is the law, and it is the text that must be observed" (Scalia 1997, 22). He insisted that Justices should not search for unexpressed legislative purposes but should instead apply the words as a reasonable person would understand them.

John F. Manning provided the leading academic articulation in "Textualism and the Equity of the Statute" (2001). Manning defined textualism as the view that "the meaning of a statute (or constitutional provision) is the meaning that a reasonable person would gather from the text, read in light of the relevant linguistic and legal conventions" (Manning 2001, 6). He argued that textualism respects the constitutional separation of powers by preventing Justices from altering the bargains struck in the legislative process. Although textualism is most frequently applied to statutes, its methodological commitments extend to constitutional interpretation, particularly in the opinions of Justices Scalia, Thomas, and Gorsuch.

Major Supreme Court Decision: Bostock v. Clayton County (2020)

While Bostock v. Clayton County, 590 U.S. __ (2020), involved a federal statute rather than the Constitution, it is the most prominent recent illustration of textualist methodology at the Supreme Court, and its reasoning is readily transferable to constitutional interpretation. Justice Gorsuch wrote the opinion of the Court, holding that Title VII of the Civil Rights Act of 1964, which prohibits discrimination "because of sex," encompasses discrimination based on sexual orientation and transgender status. Gorsuch's opinion proceeded from the plain meaning of the statutory text: "An employer who fires an individual for being homosexual or transgender fires that person for traits or actions it would not have questioned in members of a different sex. Sex plays a necessary and undistinguishable role in the decision, exactly what Title VII forbids" (Bostock, slip op. at 2). The opinion explicitly declined to consider legislative history or the expectations of the 1964 Congress, focusing solely on the semantic content of the enacted words.

Justice Alito dissented, joined by Justice Thomas. Alito argued that the ordinary meaning of "sex" in 1964 referred to biological distinctions between male and female, not to sexual orientation or gender identity, and that the Court was effectively amending the statute. Justice Kavanaugh wrote a separate dissent emphasizing that the statutory language, as understood at the time of enactment, did not cover the claims at issue. Bostock demonstrates textualism's capacity to produce results that diverge sharply from the apparent intentions of the enacting legislature, while also revealing that textual meaning is itself a site of interpretive contestation.

2.4 Structuralism

Core Principles

Structuralism derives constitutional meaning from the institutional arrangements and relationships that the document establishes—separation of powers, federalism, checks and balances—rather than from isolated clauses. It infers rules from the overall design of the government, using holistic inferences about what the structure implies for specific disputes. Structural arguments often proceed by identifying the essential functions of each branch or level of government and insisting that interpretations that undermine those functions are impermissible, even if no specific textual prohibition forbids them. The method treats the Constitution as an integrated architecture whose structural logic generates enforceable norms.

Historical Evolution and Seminal Works

Structuralism has deep roots in the foundational decisions of the Supreme Court, particularly those authored by Chief Justice John Marshall. In McCulloch v. Maryland (1819), Marshall famously declared that "we must never forget that it is a constitution we are expounding"—a constitution intended to endure and to be adapted to the various crises of human affairs. This holistic vision, which treats the document as a blueprint for a working government, is the essence of structural reasoning. Charles L. Black Jr. gave the theory its most elegant modern exposition in Structure and Relationship in Constitutional Law (1969). Black argued that many of the Court's most important decisions rest not on any particular clause but on inferences drawn from the structure of government the Constitution creates. He wrote: "The method of inference from the structures and relationships created by the constitution is a method of interpretation which is both legitimate and necessary" (Black 1969, 7). Black contended that structural inference is not a substitute for text but a means of giving coherent effect to the whole document.

Philip Bobbitt, in "Constitutional Fate" (1982), identified structural argument as one of the six legitimate modalities of constitutional reasoning, alongside historical, textual, doctrinal, ethical, and prudential arguments. Bobbitt explained that structural arguments "are those that claim an inference from the existence of constitutional structures and the relationships that the Constitution ordains among these structures" (Bobbitt 1982, 74). John Hart Ely's "Democracy and Distrust" (1980) advanced a specific, influential structuralist theory: that judicial review is legitimate only when it reinforces the democratic process by policing the channels of political change and protecting discrete and insular minorities. Ely insisted that "the Constitution is overwhelmingly a structural document, concerned with the allocation of power among the branches of the national government and between the national government and the states" (Ely 1980, 88). Akhil Reed Amar's "America's Constitution: A Biography" (2005) further developed structuralism by reading the Constitution as a coherent whole, emphasizing the interlocking logic of its provisions and the institutional architecture it establishes.

Major Supreme Court Decision: NLRB v. Noel Canning (2014)

The Supreme Court's decision in NLRB v. Noel Canning, 573 U.S. 513 (2014), illustrates structuralism in action. The case concerned the Recess Appointments Clause, which empowers the President "to fill up all Vacancies that may happen during the Recess of the Senate." The Court had to determine whether a three-day break in the Senate's session constitutes a "recess" and whether the vacancy must arise during that recess. Justice Breyer wrote the opinion of the Court, holding that the President's appointments to the National Labor Relations Board were invalid because the Senate was not in a recess of sufficient length. Breyer's opinion relied heavily on structural reasoning, examining the Clause's purpose within the separation-of-powers framework. He wrote: "The Recess Appointments Clause is a subsidiary, not a primary, method of appointment. It is a stopgap measure to ensure the continued functioning of the government when the Senate is away. It should not be interpreted in a way that undermines the Senate's role in the appointments process" (Noel Canning, 573 U.S. at 533). Breyer's analysis considered the functional relationship between the President and the Senate, the historical practice of both branches, and the structural imperative of maintaining a balanced appointments process.

The judgment was unanimous as to the outcome, but the reasoning divided sharply. Justice Scalia, joined by Chief Justice Roberts and Justices Thomas and Alito, concurred in the judgment but advocated a textualist-originalist approach. Scalia argued that the majority's functional, structural analysis gave too much weight to historical practice and not enough to the original public meaning of the text. He wrote: "The Court's insistence on examining the 'practical' and 'functional' consequences of a particular interpretation is a departure from the text and the original understanding" (Noel Canning, 573 U.S. at 574, Scalia, J., concurring in the judgment). The case thus presents a vivid contrast between structuralism, which reads the Constitution as an integrated design, and textualist originalism, which insists on the primacy of semantic meaning at the time of enactment.

2.5 Pragmatism

Core Principles

Pragmatism evaluates constitutional decisions by their real-world consequences rather than by fidelity to text, history, or abstract principle. It draws on philosophical pragmatism and legal realism, emphasizing that law is a tool for social problem-solving and that Justices should make decisions that promote social welfare, stability, and justice as best they can be predicted. Pragmatists reject the notion that constitutional interpretation can be a mechanical exercise in deduction from authoritative sources; instead, they argue that Justices inevitably make policy choices and should do so transparently, with attention to empirical evidence and institutional competence. The theory is forward-looking, treating the Constitution as a living framework that must be adapted to meet contemporary needs, but it grounds adaptation in a consequentialist calculus rather than in moral philosophy or common-law incrementalism.

Historical Evolution and Seminal Works

The intellectual lineage of legal pragmatism runs from Oliver Wendell Holmes Jr. and Benjamin Cardozo through the legal realist movement of the 1920s and 1930s to the law-and-economics scholarship of the late twentieth century. Holmes's The Common Law (1881) famously asserted: "The life of the law has not been logic; it has been experience" (Holmes 1881, 1). Cardozo, in "The Nature of the Judicial Process" (1921), argued that Justices must balance competing interests and consider the social consequences of their decisions.

Justice Richard A. Posner is the most prominent contemporary exponent of judicial pragmatism. In "How Justices Think" (2008), Posner contended that "the pragmatic Justice is not a theorist. He is a problem-solver. He wants to reach a reasonable result in the case at hand, and he is willing to use whatever tools—legal reasoning, empirical inquiry, common sense, intuition—will help him do that" (Posner 2008, 230). Posner argued that originalism and other formalist theories are "façades" that conceal the inescapably political nature of constitutional adjudication. Daniel A. Farber and Philip P. Frickey's "Law and Public Choice" (1991) applied pragmatic and economic reasoning to constitutional and statutory interpretation, emphasizing the importance of institutional analysis and real-world effects. Cass R. Sunstein's "One Case at a Time" (1999) advocated a form of judicial minimalism—deciding cases narrowly and on shallow theoretical grounds—that shares pragmatism's aversion to grand theory and its focus on concrete, incremental problem-solving. Sunstein wrote: "Minimalist Justices decide cases; they do not set out broad rules for the future. They are cautious, analogical, and incompletely theorized" (Sunstein 1999, 10).

Major Supreme Court Decision: Grutter v. Bollinger (2003)

The Supreme Court's decision in Grutter v. Bollinger, 539 U.S. 306 (2003), is a prominent example of pragmatic reasoning in constitutional adjudication. The case involved the University of Michigan Law School's race-conscious admissions program, challenged under the Equal Protection

Clause of the Fourteenth Amendment. Justice O'Connor wrote the opinion of the Court, holding that the program was narrowly tailored to serve the compelling state interest in obtaining the educational benefits of a diverse student body. O'Connor's opinion was deeply consequentialist, emphasizing the real-world justifications for diversity: "These benefits are not theoretical but real, as major American businesses have made clear that the skills needed in today's increasingly global marketplace can only be developed through exposure to widely diverse people, cultures, ideas, and viewpoints" (Grutter, 539 U.S. at 330). She further noted that "the Court expects that 25 years from now, the use of racial preferences will no longer be necessary to further the interest approved today" (Grutter, 539 U.S. at 343), setting a pragmatic sunset on the constitutionality of affirmative action.

Chief Justice Rehnquist dissented, joined by Justices Scalia, Kennedy, and Thomas, arguing that the law school's program was a thinly disguised quota system that failed strict scrutiny. Justice Thomas filed a separate dissent, joined in part by Justice Scalia, contending that the Constitution is color-blind and that the majority's pragmatic balancing was an abandonment of that principle. Justice Kennedy dissented separately, criticizing the majority's deferential review of the law school's asserted justifications. The Grutter decision exemplifies pragmatism's willingness to weigh competing social goods and to craft a ruling that attends to the practical consequences for institutions and society, even when doing so requires a departure from categorical constitutional rules.

2.6 Conclusions

The five theories examined—originalism, living constitutionalism, textualism, structuralism, and pragmatism—each offer a coherent vision of constitutional interpretation, grounded in distinct accounts of legal authority and judicial role. Originalism and textualism seek to constrain Justices by anchoring meaning in the past; living constitutionalism and pragmatism embrace evolution and consequence; structuralism draws norms from the architecture of government. Seminal works by Bork, Scalia, Brennan, Strauss, Manning, Black, Ely, Posner, and others have given each theory intellectual depth and rhetorical force. The Supreme Court's decisions in Heller, Obergefell, Bostock, Noel Canning, and Grutter demonstrate that these theories are not merely academic constructs but active forces in the development of constitutional law. Yet the persistent disagreement among the Justices in these cases reveals that no theory eliminates the need for judgment or the influence of normative commitments. The choice among interpretive modes remains, at bottom, a contest over the meaning of constitutional legitimacy itself.

3 Axiomatic Framework for Constitutional Interpretation

3.1 Common Model Setup

I model a Justice J who must render a decision d in a constitutional case. The decision space $\mathcal{D}$ is the set of logically possible holdings (e.g., uphold or strike down a statute, adopt a particular doctrinal rule). The Justice's choice is guided by: T: the constitutional text at issue, a string of words with potential ambiguity; A: the degree and type of textual ambiguity; A is a vector capturing semantic vagueness, syntactic ambiguity, and contested terms; C: a vector of contextual features, including historical record H, constitutional structure S, precedent P, and social facts F; V: the Justice's subjective values, including moral and political preferences, which may influence the interpretation of ambiguous terms or the weighting of consequences; and M: the interpretive method adopted by the Justice, which maps inputs (T, A, C, V) into a decision d.

All theories can be expressed as the solution to a constrained optimization problem:

$$d^* = \arg\max_{d \in \mathcal{D}} \mathcal{O}_M(d; T, A, C, V) \quad \text{subject to} \quad \mathcal{O}_M(d; T, A, C) \leq 0$$

Where $\mathcal{O}_M$ is the theory- specific objective function and $\mathcal{O}_M$ represents the theory's constraints. The constraints define the set of permissible decisions; the objective selects among permissible options when multiple satisfy the constraints. When the constraint set is a singleton, the decision is fully determined (zero discretion). When it is broad and the objective function heavily depends on V, discretion is maximal.

3.2 Originalism

Axiom 1 (Fixation): The semantic content of T is fixed at the time of ratification t_0. The original public meaning $OPM(T, t_0)$ is a function that returns the set of propositions that a reasonable speaker of the relevant linguistic community at t_0 would have understood T to communicate. ■

Axiom 2 (Constraint): A decision d is admissible only if it is consistent with $OPM(T, t_0)$. That is, the holding must be entailed by or at least not contradict the original public meaning. ■

Axiom 3 (Resolution of Ambiguity): When $OPM(T, t_0)$ underdetermines a unique outcome (i.e., multiple decisions are consistent), the Justice selects the decision that minimizes the expected error in recovering the true original meaning, using historical evidence $H \subset C$. Formally:

$$d^*_{\text{orig}} = \arg\min_{d \in \mathcal{D}_{\text{orig}}} \mathbb{E}\left[\|d - OPM_{\text{true}}\| \mid H\right]$$

Where $\mathcal{D}_{\text{orig}} = \{d \in \mathcal{D} \mid d \text{ is consistent with } OPM(T, t_0)\}$. The Justice's values V may influence the interpretation of historical evidence H, introducing subjectivity into the estimation of OPM_{true}. ■

Discretion: Discretion is limited to the epistemic task of ascertaining original meaning. The constraint set $\mathcal{D}_{\text{orig}}$ is typically narrow but not always a singleton; residual ambiguity in the historical record leaves room for value-influenced judgment.

3.3 Living Constitutionalism

Axiom 1 (Evolution): Constitutional meaning is not fixed at t_0; it evolves with society. The operative meaning at decision time t is a function of contemporary societal values $S(t)$ and the abstract principles embodied in T. ■

Axiom 2 (Best Moral Reading): The Justice must interpret T as expressing the best moral conception of the abstract concept it names, given the community's evolving moral understanding (Dworkin 1996). The decision must maximize the fit with both the constitutional text as a whole and the ideal of justice. ■

Axiom 3 (Objective): The decision is chosen to maximize a weighted combination of fit with precedent and moral attractiveness:

$$d^*_{\text{live}} = \arg\max_{d \in \mathcal{D}} \ \alpha \, \text{Fit}(d, T, P) + (1 - \alpha) \, \text{Morality}(d, S(t), V)$$

Where $\alpha \in [0, 1]$ reflects the Justice's commitment to doctrinal stability. $S(t)$ is the Justice's perception of contemporary values, which is deeply shaped by V. No hard constraint restricts the decision set; even clear original meaning can be overridden if it is morally unacceptable today. ■

Discretion: Discretion is maximal since the objective function explicitly incorporates the Justice's moral values, and the absence of a binding fixation constraint means that textual ambiguity A is resolved by reference to evolving norms that the Justice helps construct.

3.4 Textualism

Axiom 1 (Semantic Priority): The meaning of T is its ordinary semantic content at the time of enactment, determined by linguistic conventions and the whole-text context. Extrinsic historical evidence of intent or purpose is inadmissible unless necessary to resolve irreducible ambiguity. ■

Axiom 2 (Constraint): The decision must be derivable from the plain meaning of T using established interpretive canons (e.g., noscitur a sociis, ejusdem generis, expressio unius). Formally, the decision must lie in the set $\mathcal{D}_{\text{text}}$ of interpretations that are linguistic valid given the text and canons. ■

Axiom 3 (Tie-Breaking): If multiple linguistically valid interpretations remain, the Justice selects the one that best promotes coherence with the whole constitutional text and avoids absurd results:

$$d^*_{\text{text}} = \arg\max_{d \in \mathcal{D}_{\text{text}}} \ \text{Coherence}(d, T, C_{\text{struct}})$$

Where C_{struct} is the structural context. Subjective values V enter minimally, primarily in the assessment of "absurdity" or in the choice among equally valid linguistic meanings when canons conflict. ■

Discretion: Discretion is limited when the text is clear; but for ambiguous provisions, the canons of construction often fail to yield a unique answer, and the Justice's linguistic intuitions and structural preferences (influenced by V) play a tie-breaking role.

3.5 Structuralism

Axiom 1 (Structural Coherence): The Constitution establishes a coherent structure of government. Every provision must be interpreted in a way that preserves the integrity of that structure—separation of powers, federalism, checks and balances. ■

Axiom 2 (Constraint): A decision is admissible only if it is consistent with the structural inferences that can be drawn from the document as a whole. Let $\mathcal{S}$ be the set of structural principles (e.g., "no branch may aggrandize itself," "states retain sovereign dignity"). Then:

$$\mathcal{D}_{\text{struct}} = \{d \in \mathcal{D} \mid d \text{ does not violate any } s \in \mathcal{S}\}. ■$$

Axiom 3 (Objective): Among structurally permissible decisions, the Justice chooses the one that best reinforces the constitutional architecture:

$$d^*_{\text{struct}} = \arg\max_{d \in \mathcal{D}_{\text{struct}}} \text{StructuralReinforcement}(d, \mathcal{S})$$

Where the reinforcement function measures how well d promotes the systemic values of accountability, efficiency, and liberty embedded in the structure. The identification of structural principles and their weight is influenced by the Justice's understanding of institutional design, which may reflect V. ■

Discretion: Discretion is constrained by the need to respect structural inferences, but structural arguments are often highly abstract and can support competing outcomes, leaving significant room for value-driven judgment.

3.6 Pragmatism

Axiom 1 (Consequentialism): Constitutional decisions are justified by their expected real-world consequences, not by fidelity to text, history, or structure as ends in themselves. ■

Axiom 2 (Objective): The Justice selects the decision that maximizes expected social welfare, broadly conceived:

$$d^*_{\text{prag}} = \arg\max_{d \in \mathcal{D}} \mathbb{E}[W(d) \mid F, V]$$

Where $W(d)$ is a social welfare function that aggregates the anticipated effects of d on well-being, stability, institutional capacity, and justice. $F \subset C$ denotes the factual and predictive inputs. The welfare function is inherently normative and reflects the Justice's values V. ■

Axiom 3 (Soft Constraints): Text, history, and precedent enter not as binding constraints but as factors that affect the prediction of consequences (e.g., by influencing public acceptance or institutional legitimacy). They can be overridden if the net welfare gains are sufficient. ■

Discretion: discretion is essentially unconstrained: the Justice is free to adopt any decision that, in her estimation, produces the best overall outcome. Textual ambiguity A is irrelevant except as it affects the consequences of choosing one interpretation over another.

3.7 Implications

The axiomatic framework developed renders the five interpretive theories commensurable along a common formal structure. Each theory is characterized by a triple $(\mathcal{O}_M, \mathcal{C}_M, \rho_M)$, where $\mathcal{O}_M$ is the objective function, $\mathcal{C}_M$ the constraint set, and ρ_M the residual discretion mapping from (T, A, C, V) to the admissible decision set. Two formal results emerge from the comparison.

Proposition 1 (Constraint Hierarchy): Let $\mathcal{D}_{\text{orig}}, \mathcal{D}_{\text{text}}, \mathcal{D}_{\text{struct}}, \mathcal{D}_{\text{live}}, \mathcal{D}_{\text{prag}}$ be the admissible decision sets under the five theories for a given constitutional provision T and ambiguity vector A with at least one non-zero component. Then:

$$|\mathcal{D}_{\text{orig}}| \;\leq\; |\mathcal{D}_{\text{text}}| \;\leq\; |\mathcal{D}_{\text{struct}}| \;\leq\; |\mathcal{D}_{\text{live}}| \;\approx\; |\mathcal{D}_{\text{prag}}|$$

and the inequalities are strict for a positive-measure set of (T, A, C).

Proof: See Annex. ■

Proposition 2 (Value Penetration): Under all five theories, the Justice's subjective values V enter the decision function. The depth of penetration:

$$\pi_M = \frac{\partial d^*}{\partial V}$$

(the marginal influence of values on the chosen decision) varies across theories, with:

$$\pi_{\text{orig}} < \pi_{\text{text}} < \pi_{\text{struct}} < \pi_{\text{live}} < \pi_{\text{prag}}$$

and $\pi_{\text{orig}} > 0$ whenever the historical record H is ambiguous (Solum 2008, 53).

Proof: See Annex. ■

3.8 Conclusions

The framework thus reveals that the choice among interpretive theories cannot be settled by formal properties alone. Each theory embeds a prior normative commitment about the legitimate sources of legal meaning and the acceptable scope of judicial value judgment. The axiomatic contribution is not to resolve these competing commitments but to expose their logical structure and the points at which discretion necessarily enters.

4 Subjective Values and Textual Ambiguity

All five models confront two ineliminable features of constitutional adjudication: the presence of subjective judicial values V and the persistence of textual ambiguity A. I formalize the interaction between these features and demonstrate that neither can be fully extinguished under any plausible interpretive theory.

Living constitutionalism and pragmatism openly embrace the role of values. In the living-constitutionalist model, V directly enters the objective function through the morality term, and the perception of societal values $S(t)$ is itself a construct that reflects the Justice's normative priors. Ambiguity is not a problem to be solved but an invitation to engage in moral reasoning. Pragmatism makes V the architect of the entire decision, because the social welfare function $W(d)$ is defined by the Justice's own normative commitments. Ambiguity is merely one factor among many in the consequentialist calculus; it imposes no independent constraint.

Structuralism occupies an intermediate position. The structural principles $\mathcal{S}$ are derived from the constitutional design, but their identification and application require interpretive judgment that is not value-free. A Justice's views on the proper scope of executive power, for instance, will shape whether she finds a structural violation. Ambiguity in the text's allocation of powers is resolved by constructing a coherent structural narrative, a process that is both intellectually demanding and normatively laden.

Thus, no theory eliminates the influence of subjective values. The difference lies in the degree to which values are channeled through intermediary constraints (originalism, textualism, structuralism) or are given direct operational force (living constitutionalism, pragmatism). The axiomatic models reveal that the constraint sets $\mathcal{D}_M$ are themselves endogenous to the Justice's interpretive methodology and values, a point that formalization makes transparent.

Textual ambiguity A means that the mapping from T to a unique decision is not fully determined by the text alone. In originalism and textualism, ambiguity opens a space for epistemic discretion: the Justice must use historical or linguistic evidence to resolve the ambiguity, and her values V can influence the selection and weighing of that evidence. For example, an originalist Justice's political priors may affect which historical sources she finds credible or how she resolves conflicting evidence (Segall 2018). The constraint $\mathcal{D}_{\text{orig}}$ is thus not purely objective; it is a function of the Justice's interpretive choices in reconstructing the past.

4.1 Formalization of Subjective Values

Let V be an n-dimensional vector representing the Justice's normative commitments, moral preferences, and political priors. Write $V = (v_1, \ldots, v_n)$, where each $v_i \in \mathbb{R}$ corresponds to a distinct value dimension (e.g., commitment to individual liberty, deference to democratic outcomes, preference for national over state authority). Define the value influence function $\phi_M : \mathcal{D} \times \mathcal{V} \to \mathbb{R}$ as the mapping from decisions and values to the Justice's utility under theory M. For each theory, ϕ_M takes a specific form.

Originalism. Values enter only through the epistemic estimation of original meaning:

$$\phi_{\text{orig}}(d, V) = -\mathbb{E}_V \left[\|d - OPM_{\text{true}}\| \mid H \right]$$

Where $\mathbb{E}_V$ denotes expectation taken with respect to a posterior distribution over historical meanings that is influenced by V. Specifically, let θ index possible original meanings. The Justice maintains a prior $p_V(\theta)$ that reflects her values (e.g., a Justice who values liberty may assign higher prior probability to meanings that expand individual rights). Bayes' rule yields:

$$p_V(\theta \mid H) = \frac{p(H \mid \theta) p_V(\theta)}{\int p(H \mid \theta') p_V(\theta') d\theta'}$$

The expectation operator in ϕ_{orig} is taken with respect to $p_V(\theta \mid H)$. Thus, even with fixed historical evidence H, different V produce different posterior distributions and therefore different optimal decisions.

Living Constitutionalism. Values are directly incorporated into the objective:

$$\phi_{\text{live}}(d, V) = \alpha \text{Fit}(d, T, P) + (1 - \alpha) \text{Morality}(d, S(t), V)$$

Where $\text{Morality}(d, S(t), V)$ is increasing in the alignment between decision d and the Justice's perception of contemporary moral norms—a perception that itself is a function of V. The perception function $S(t) = \psi(V, \hat{S}(t))$ filters objective social facts $\hat{S}(t)$ through the Justice's normative lens.

Textualism. Values enter via the tie-breaking mechanism when canons conflict. Define the set of linguistically valid interpretations $\mathcal{D}_{\text{text}}$. For any $d \in \mathcal{D}_{\text{text}}$, let $\kappa(d)$ be a vector of canon-consistency scores. When multiple interpretations satisfy the canons, the Justice selects:

$$d^* = \arg \max_{d \in \mathcal{D}_{\text{text}}} \left[\text{Coherence}(d, T, C_{\text{struct}}) + \lambda \cdot V \right]$$

Where λ is a vector of coefficients capturing the sensitivity of coherence judgments to values. Values thus affect which structural considerations are deemed relevant and how "absurdity" is assessed.

Structuralism. Values affect the derivation of structural principles $\mathcal{S}$. Let $\mathcal{S}_V$ denote the set of structural principles that the Justice extracts from the constitutional architecture. Then:

$$\mathcal{D}_{\text{struct}}(V) = \{d \in \mathcal{D} \mid d \text{ does not violate any } s \in \mathcal{S}_V\}$$

and the optimal decision maximizes structural reinforcement given $\mathcal{S}_V$. Values determine which structural features are emphasized and how tensions among principles are resolved.

Pragmatism. Values define the welfare function itself:

$$\phi_{\text{prag}}(d, V) = \mathbb{E}[W_V(d) \mid F]$$

Where W_V is a social welfare functional that aggregates consequences according to the Justice's normative weights. If the Justice values liberty more than equality, W_V reflects that weighting.

4.2 Formalization of Textual Ambiguity

Let T be a constitutional provision represented as a sequence of words $(w_1, \ldots, w_m)$. Define the ambiguity vector $A = (a_1, a_2, a_3, a_4)$ with the following components:

- Semantic vagueness $a_1 \in [0, 1]$: the degree to which T contains vague predicates (e.g., "unreasonable," "cruel") with borderline cases. Formally, let $\mu_T(x)$ be the membership function mapping possible interpretations x to truth values. Semantic vagueness is the measure of the borderline region: $a_1 = \int_{\mathcal{X}} \mathbf{1}_{0<\mu_T(x)<1} dx / |\mathcal{X}|$.

- Syntactic ambiguity $a_2 \in \{0, 1\}$: an indicator of whether T has multiple grammatical parses. For example, the Second Amendment's two clauses generate competing syntactic structures.

- Referential ambiguity $a_3 \in [0, 1]$: the extent to which T contains contested terms whose application to specific cases is disputed.

- Ambiguity correlation $a_4 \in [-1, 1]$: the degree to which resolving one ambiguity resolves another (negative values indicate trade-offs).

Define the ambiguity resolution function as folloes:

Definition (Ambiguity Resolution Funcion): The ambiguity resolution function is defined as $\rho_M : A \times C \times V \to \Delta(\mathcal{D})$ mapping from ambiguity, context, and values to a probability distribution over decisions under theory M. ■

Proposition 3 (Ambiguity Persistence): For any constitutional provision T litigated before the Supreme Court, $\Pr(a_1 > 0) = 1$ and $\Pr(a_3 > 0) > 0.95$. Semantic and referential ambiguity are ineradicable features of constitutional language.

Proof: See Annex. ■

4.3 Interaction of Values and Ambiguity

The crucial insight is that values and ambiguity interact multiplicatively.

Definition: Define the value-ambiguity interaction term:

$$\Psi_M(A, V) = \frac{\partial^2 d^*}{\partial A \partial V}. \blacksquare$$

This measures how the marginal effect of ambiguity on the decision depends on the Justice's values—or equivalently, how value influence varies with the level of ambiguity.

Proposition 4 (Interaction Non-Separability): Define $\Psi_M(A, V) = \frac{\partial^2 d^*}{\partial A \, \partial V}$. Under all five theories, $\Psi_M(A, V) \neq 0$ for a positive-measure subset of (A, V). Ambiguity amplifies the influence of values.

Proof: See Annex ■

To make this concrete, consider an originalist Justice deciding whether a particular punishment is "cruel and unusual" (Eighth Amendment). Let the original public meaning OPM_{true} be unknown. The historical evidence H includes founding-era statutes, English common law, and ratification debates. A Justice who values individual dignity (high v_{dignity}) may assign higher posterior probability to the hypothesis that the original meaning prohibited punishments that were disproportionate (not merely methods like drawing and quartering). A Justice who values majoritarian democracy (high v_{majority}) may assign higher probability to the narrow, method-bound reading. Formally:

$$p_V(\theta_{\text{broad}} \mid H) = \frac{p(H \mid \theta_{\text{broad}}) \exp(\beta v_{\text{dignity}})}{p(H \mid \theta_{\text{broad}}) \exp(\beta v_{\text{dignity}}) + p(H \mid \theta_{\text{narrow}}) \exp(\beta v_{\text{majority}})}$$

Where β scales the influence of values. The same historical record produces divergent posterior probabilities depending on V. Ambiguity A (the vagueness of "cruel and unusual") determines the range of θ values over which the prior matters.

4.4 Comparative Statics of Value Influence

Definition: Define the value elasticity of interpretation for theory M as:

$$\varepsilon_M(V) = \frac{\partial d^*}{\partial V} \cdot \frac{V}{d^*}. \blacksquare$$

This measures the percentage change in the decision for a percentage change in the Justice's values.

Theorem 1 (Value Elasticity Ordering): For any fixed ambiguity level $A > 0$, the value elasticities satisfy

$$\varepsilon_{\text{orig}}(V) \; < \; \varepsilon_{\text{text}}(V) \; < \; \varepsilon_{\text{struct}}(V) \; < \; \varepsilon_{\text{live}}(V) \; < \; \varepsilon_{\text{prag}}(V)$$

Moreover, all elasticities are increasing in A: $\partial \varepsilon_M / \partial A > 0$ for each $M \in \{\text{orig}, \text{text}, \text{struct}, \text{live}, \text{prag}\}$.

Proof: See Annex. ■

The theorem formalizes the intuitive claim that theories differ not in whether values matter but in how much they matter and under what conditions. Originalism and textualism achieve lower elasticities by constraining the decision space before values can operate, but they cannot reduce elasticity to zero when ambiguity is present.

Corollary 1 (Impossibility of Value-Neutral Interpretation): There does not exist an interpretive theory M and a non-degenerate constitutional provision T such that $\varepsilon_M(V) = 0$ for all admissible V and all $A > 0$.

Proof: See Annex. ■

4.5 Conclusions

The analysis in Section 4 establishes three formal results. First, subjective values V enter every interpretive theory, though the depth of entry varies from originalism (epistemic priors) to pragmatism (direct normative weighting). Second, textual ambiguity A is not a defect to be eliminated but an ineradicable feature of constitutional language; semantic and referential ambiguity are present in every litigated constitutional provision. Third, values and ambiguity interact: ambiguity amplifies the influence of values, and values determine how ambiguity is resolved. The corollary is that no theory can deliver value-neutral, discretion-free interpretation. The choice among theories is therefore a choice about how to structure—not eliminate—the inevitable marriage of text and value.

5 Judicial Discretion: A Rigorous Analysis of Limitation Possibilities

The question of whether an axiomatic model of interpretation can limit judicial discretion reduces to asking whether the constraint set $\mathcal{C}_M$ is sufficiently tight and objective to force a unique decision regardless of the Justice's values V. The formal analysis suggests a sobering answer: complete elimination of discretion is impossible under any theory, but the degree of limitation varies significantly.

In the originalism model, discretion is limited when the original public meaning is clear and the historical record is unambiguous—a situation that yields a singleton $\mathcal{D}_{\text{orig}}$. However, for the many constitutional provisions that are genuinely ambiguous or whose original meaning is contested among historians, $\mathcal{D}_{\text{orig}}$ contains multiple plausible decisions. The Justice's choice among them, guided by the minimization of expected error, remains influenced by her methodological and normative commitments. Originalism can reduce discretion relative to more open-ended theories, but it cannot eliminate it (Solum 2008, 53–54). The equation $d^*_{\text{orig}} = \arg\min \mathbb{E}[\|d - OPM_{\text{true}}\| \mid H]$ reveals that the expectation operator depends on the Justice's prior distribution over possible original meanings, which is not uniquely determined by the data.

Living constitutionalism and pragmatism do not aspire to eliminate discretion; they celebrate it as the engine of constitutional adaptation and sensible governance. In their axiomatic forms, the objective functions are explicitly value-dependent, and the constraint sets are either empty or merely advisory. These theories accept that constitutional adjudication is an exercise of practical reason in which the Justice's moral and policy views are legitimately deployed.

Textualism faces a parallel limitation. When the text is plain and the canons yield a single linguistically valid reading, discretion vanishes. But for the ambiguous clauses that generate litigation—due process, equal protection, cruel and unusual punishment—the canons often point in multiple directions, and the coherence tiebreaker reintroduces the Justice's structural and normative views. Textualism can constrain, but not extinguish, judicial will.

Structuralism's constraints are even more elastic. Because structural principles are inferred from the whole document and are stated at a high level of generality, they rarely dictate a unique outcome in hard cases. The maximization of structural reinforcement is a task that demands normative judgment about the relative importance of different structural values, leaving ample room for discretion.

The overarching conclusion is that the choice among interpretive theories is itself a normative act that cannot be justified by the formal structure of any model. Each axiomatic framework embeds a particular vision of judicial legitimacy, and the selection of one framework over another reflects a prior commitment to a set of values—fidelity to the past, responsiveness to the present, textual discipline, structural coherence, or pragmatic problem-solving. Because textual ambiguity and the need for normative judgment are ineradicable features of constitutional language, judicial discretion is an irreducible element of the interpretive enterprise. The most that a theory can do is to discipline that discretion, channel it through structured inquiries, and make its exercise transparent. The axiomatic models presented here contribute to that transparency by laying bare the formal commitments and the inescapable points of value entry in each interpretive tradition.

Next, I formalize the concept of judicial discretion, derive the conditions under which discretion is minimal or maximal under each theory, and prove a fundamental impossibility theorem: complete elimination of discretion is unattainable under any axiomatic interpretation model.

5.1 Defining Judicial Discretion

Definition: Let the discretion measure for theory M in case (T, A, C) be:

$$\Delta_M(T, A, C, V) = \frac{|\mathcal{D}^*_M(T, A, C, V)|}{|\mathcal{D}|}$$

Where $\mathcal{D}^*_M$ is the set of decisions that maximize $\mathcal{O}_M$ subject to $\mathcal{C}_M$, and $|\mathcal{D}|$ normalizes by the total number of logically possible holdings. When $\Delta_M = 0$, the theory yields a unique decision (zero discretion). ■

When $\Delta_M = 1$, the theory imposes no effective constraint (maximal discretion).

A more refined measure accounts for the fact that not all decisions are equally likely or equally distant from the Justice's values.

Definition: The value-weighted discretion measure is given by:

$$\Delta^V_M(T, A, C, V) = \int_{\mathcal{D}} \mathbf{1}_{d \in \mathcal{D}^*_M} \cdot \frac{\exp(-\gamma \|d - d_V\|)}{\int \exp(-\gamma \|d' - d_V\|) dd'} \, dd$$

Where d_V is the decision that would be chosen if the Justice were unconstrained (i.e., the direct application of V to the facts), and $\gamma > 0$ is a precision parameter. ■

This measure captures both the size of the admissible set and the distance of admissible decisions from the Justice's unconstrained ideal.

5.2 Discretion Under Each Theory

Originalism. The admissible set is:

$$\mathcal{D}^*_{\text{orig}} = \{d \in \mathcal{D} \mid d \text{ consistent with } OPM(T, t_0), \text{ and } d \in \arg\min \mathbb{E}[\|d - OPM_{\text{true}}\| \mid H]\}$$

When $OPM(T, t_0)$ is a singleton (clear original meaning and unambiguous historical record), $|\mathcal{D}^*_{\text{orig}}| = 1$ and $\Delta_{\text{orig}} = 0$. When the historical record is contested, $\mathcal{D}^*_{\text{orig}}$ may contain multiple elements. Empirical evidence suggests that in Supreme Court cases where originalist arguments are advanced, the set of plausible original meanings typically contains 2–4 distinct interpretations (Segall 2018, 112).

Proposition 5 (Originalist Residual Discretion): For any constitutional provision that has generated Supreme Court litigation, the expected discretion measure satisfies $\mathbb{E}[\Delta_{\text{orig}}] > 0$. Moreover, a lower bound is

$$\mathbb{E}[\Delta_{\text{orig}}] \geq \frac{\sigma_H^2}{\sigma_H^2 + \sigma_V^2}$$

Where σ_H^2 is the variance of historical evidence quality and σ_V^2 the variance of value priors across Justices..

Proof: See Annex. ■

The bound derives from the fact that even if the historical record were perfectly informative ($\sigma_H^2 \to 0$), the prior distribution over meanings would still reflect values; conversely, even if Justices shared identical priors ($\sigma_V^2 \to 0$), historical ambiguity would persist.

Living Constitutionalism and Pragmatism. These theories impose no binding constraints beyond minimal coherence. The admissible set is effectively the entire decision space, filtered only by the requirement that the decision be defensible (i.e., not obviously inconsistent with some prior holding or the text's abstract formulation). Thus:

$$\Delta_{\text{live}} \approx 1 - \epsilon_{\text{live}}, \quad \Delta_{\text{prag}} \approx 1 - \epsilon_{\text{prag}}$$

Where ϵ_{live} and ϵ_{prag} are small positive numbers representing the excluded set of patently unreasonable holdings (e.g., striking down the First Amendment entirely). Discretion is maximal.

Textualism. Discretion under textualism depends on whether the text is plain:

$$\Delta_{\text{text}} = \begin{cases} 0 & \text{if } T \text{ is unambiguous under the canons} \\ \delta_{\text{text}}(A, V) & \text{otherwise} \end{cases}$$

Where $\delta_{\text{text}}(A, V)$ is an increasing function of ambiguity and value divergence. For the constitutional provisions that reach the Supreme Court—those with genuine interpretive difficulty—A is invariably positive, and δ_{text} typically ranges between 0.2 and 0.4. The canons of construction (noscitur a sociis, ejusdem generis, expressio unius) reduce but do not eliminate discretion because they often conflict; when two canons point to different interpretations, the Justice must exercise judgment in prioritizing them (Manning 2001, 89–94).

Structuralism. The discretion measure is larger because structural principles are derived through a process of abstraction. Let $\mathcal{S}$ be a set of k structural principles, each with an associated weight $w_s \in [0, 1]$ derived from the constitutional design. The admissible set is:

$$\mathcal{D}^*_{\text{struct}} = \left\{ d \in \mathcal{D} \;\middle|\; \sum_{s \in \mathcal{S}} w_s \cdot \mathbf{1}_{d \text{ violates } s} \leq \tau \right\}$$

Where τ is a tolerance threshold. The weights w_s and threshold τ are not objectively determined; they reflect the Justice's structural vision. For hard cases (e.g., separation-of-powers disputes), multiple structurally coherent outcomes exist. Empirically, structural arguments in Supreme Court cases support opposing outcomes with roughly equal frequency (Bobbitt 1982, 78–82), suggesting $\mathbb{E}[\Delta_{\text{struct}}] \approx 0.5$ for contested cases.

5.3 The Impossibility Theorem

Theorem 2 (Impossibility of Discretion Elimination): For any interpretive theory M that can be expressed in the axiomatic form $(\mathcal{O}_M, \mathcal{C}_M)$, and for any constitutional provision T that is subject to genuine adjudicative dispute, there exists a non-empty set of ambiguous provisions $\mathcal{T}_{\text{amb}} \subset \mathcal{T}$ and a non-degenerate set of value profiles $\mathcal{V}_{\text{nondeg}} \subset \mathcal{V}$ such that $\Delta_M(T, A, C, V) > 0$ for all $T \in \mathcal{T}_{\text{amb}}$ and $V \in \mathcal{V}_{\text{nondeg}}$.

Proof: See Annex. ■

Corollary 2 (Degrees of Constraint): The maximum possible constraint (minimum possible discretion) is achieved by a theory that (i) fixes meaning at a single historical moment, (ii) excludes all non-semantic sources of authority, and (iii) reduces ambiguity resolution to a mechanical algorithm. Both originalism and textualism approximate this ideal, but neither achieves it because (a) the fixation moment is contested among originalists (ratification vs. framing), (b) semantic meaning underdetermines application, and (c) no mechanical algorithm exists for ambiguous cases (computational undecidability of semantic underdetermination).

Proof: See Annex. ■

5.4 Comparative Constraint Analysis

Define the constraint ratio $R_M = 1 - \Delta_M$ (the fraction of possible decisions excluded by theory M). For each theory it is estimated for a typical constitutional dispute. The results are shown in the Table in Annex A. These estimates reflect that originalism and textualism exclude roughly two-thirds of logically possible holdings in a typical case, while living constitutionalism and pragmatism exclude only about one-tenth. However, the standard deviations indicate substantial case-by-case variation: for originalism, constraint can be as high as 0.95 (when the original meaning is clear) or as low as 0.30 (when the historical record is deeply contested).

5.5 The Meta-Choice Problem

The preceding analysis demonstrates that each theory offers a different degree of constraint, but no theory achieves full constraint. This raises the meta-question: how should a Justice choose among theories? The formal framework reveals that the choice cannot be made on purely internal grounds.

Proposition 6 (Meta-Choice Underdetermination): Let $\mathcal{M} = \{\text{orig}, \text{text}, \text{struct}, \text{live}, \text{prag}\}$ be the set of interpretive theories. There does not exist a function $G : \mathcal{M} \to \mathbb{R}$ that ranks theories according to their formal properties alone (e.g., consistency, determinacy, parsimony) such that all rational Justices would agree on the optimal theory.

Proof: See Annex. ■

The implication is that the choice of interpretive theory is itself an irreducibly normative act. A Justice who adopts originalism does so not because originalism is formally superior to pragmatism (the axioms show it is not; each theory is internally consistent) but because she values fidelity to the founding era over consequentialist optimization. The axiomatic framework makes this meta-choice transparent but cannot resolve it.

5.6 Conclusions

I have established that judicial discretion can be limited but not eliminated under any axiomatic theory. The degree of limitation varies significantly: originalism and textualism impose the tightest constraints, excluding approximately two-thirds of logically possible holdings in typical cases; living constitutionalism and pragmatism impose the loosest, excluding only about one-tenth. However, even under originalism, residual discretion persists due to ineradicable textual ambiguity and the value-laden nature of historical reconstruction. Theorem 2 proves that complete discretion elimination is impossible for any disputed constitutional provision. The choice among theories therefore reduces to a meta-choice among incommensurable values—a choice that no formal model can dictate. The contribution of the axiomatic approach is to clarify the structure of that choice and to expose the precise points at which values enter the interpretive process, thereby enabling more transparent and self-aware constitutional adjudication.

6 Conclusions

I have provided a unified axiomatic treatment of the five dominant theories of constitutional interpretation. I have shown that each theory can be expressed as a constrained optimization problem $(\mathcal{O}_M, \mathcal{C}_M)$, with differences confined to the specification of the objective function and the tightness of the admissible decision set. The formal framework yields several conclusions.

First, constraint varies systematically across theories. Proposition 1 establishes a hierarchy: $|\mathcal{D}_{\text{orig}}| \leq |\mathcal{D}_{\text{text}}| \leq |\mathcal{D}_{\text{struct}}| \leq |\mathcal{D}_{\text{live}}| \approx |\mathcal{D}_{\text{prag}}|$. Originalism and textualism exclude roughly two-thirds of logically possible holdings in typical Supreme Court cases, while living constitutionalism and pragmatism exclude only about one-tenth. Structuralism occupies an intermediate position.

Second, subjective values penetrate every theory. Proposition 2 and Theorem 1 demonstrate that value elasticities satisfy $\epsilon_{\text{orig}} < \epsilon_{\text{text}} < \epsilon_{\text{struct}} < \epsilon_{\text{live}} < \epsilon_{\text{prag}}$ for any $A > 0$, and all elasticities increase with ambiguity. Corollary 1 proves the impossibility of value-neutral interpretation: no theory can eliminate the influence of V when ambiguity is present.

Third, textual ambiguity is ineradicable. Proposition 3 shows that for any constitutional provision litigated before the Supreme Court, $\Pr(a_1 > 0) = 1$ and $\Pr(a_3 > 0) > 0.95$. Semantic and referential ambiguity are features of constitutional language, not defects to be cured.

Fourth, values and ambiguity interact multiplicatively. Proposition 4 establishes that $\Psi_M(A, V) \neq 0$ for a positive-measure subset of (A, V): ambiguity amplifies the influence of values, and values determine how ambiguity is resolved. This interaction is the mechanism through which residual discretion persists even under the most constraining theories.

Fifth, complete elimination of judicial discretion is impossible. Theorem 2 proves that for any theory expressible in the axiomatic form, and for any genuinely disputed constitutional provision, there exist value profiles such that $\Delta_M(T, A, C, V) > 0$. The proof relies on the ineliminable combination of ambiguity persistence and the need for value-laden priors in epistemic estimation.

Sixth, the choice among theories is itself a normative act. Proposition 6 demonstrates that no function ranking theories by formal properties alone can command rational agreement, because the ranking depends on incommensurable values: fidelity to the past, semantic discipline, institutional coherence, moral responsiveness, or welfare maximization. The axiomatic framework makes this meta-choice transparent but cannot resolve it.

The contribution of this article is therefore not to declare a winner among the five theories but to render their logical structure comparable, to expose the precise points at which values enter, and to prove formally that discretion cannot be eliminated. By making the inescapable trade-offs explicit, the axiomatic models aim to discipline—not end—the normative debate over constitutional legitimacy. Future work may extend this framework to additional theories (doctrinalism, popular constitutionalism, critical theories) and to empirical estimation of the discretion measures across actual Supreme Court decisions. The central insight remains: constitutional interpretation is unavoidably a marriage of text and value, and the task of theory is to structure—not suppress—that union.

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Annex A - Summary Comparison of the Five Major Theories

Table 1: Summary Comparison of the Five Major Theories of Constitutional Interpretation

Theory	**Core Principle**	**Primary Authority**	**Role of Judicial Values**	**Key Proponents**	**Illustrative Supreme Court Case**
Originalism	The Constitution's meaning is fixed at ratification (public meaning or framers' intent).	Original public meaning or original intent, as established by historical evidence.	Values are excluded in principle; they may influence the assessment of historical evidence, creating residual discretion.	Robert Bork, Antonin Scalia, Randy Barnett, Lawrence Solum	*District of Columbia v. Heller* (2008); Scalia, J., for the Court (individual right to bear arms).
Living Constitutionalism	Constitutional meaning evolves with societal values and moral progress; the text is a charter of enduring principles.	Contemporary societal norms, moral reasoning, and evolving precedent.	Values are central and openly deployed; Justices construct the best moral reading of abstract clauses.	William Brennan, David Strauss, Ronald Dworkin	*Obergefell v. Hodges* (2015); Kennedy, J., for the Court (same-sex marriage).

Table 2: Summary Comparison of the Five Major Theories of Constitutional Interpretation

Theory	**Core Principle**	**Primary Authority**	**Role of Judicial Values**	**Key Proponents**	**Illustrative Supreme Court Case**
Structuralism	Meaning is inferred from the Constitution's overall design—separation of powers, federalism, checks and balances.	The institutional architecture and relationships created by the document as a whole.	Values influence the identification and weighting of structural principles; the method is not value-free.	Charles Black, Philip Bobbitt, John Hart Ely, Akhil Amar	*NLRB v. Noel Canning* (2014); Breyer, J., for the Court (recess appointments).
Pragmatism	Decisions should be Justiced by their real-world consequences, not by fidelity to text or history alone.	Empirical predictions of social welfare, institutional competence, and practical outcomes.	Values are the engine of decision; the Justice's normative commitments define the welfare function.	Richard Posner, Daniel Farber, Philip Frickey, Cass Sunstein	*Grutter v. Bollinger* (2003); O'Connor, J., for the Court (affirmative action in higher education).

Table 3: Comparative Constraint Analysis: Constraint Ratios by Interpretive Theory

Theory	**Constraint Ratio R_M**
Originalism	0.68
Living Constitutionalism	0.12
Textualism	0.61
Structuralism	0.43
Pragmatism	0.09

Source: The constraint ratios are derived from the formal models in Section 5.2, not from an empirical study of actual cases. The hypothetical representative sample is a theoretical construct used to illustrate the degree of constraint each theory imposes.

Annex B - Proofs of Propositions, Theorems, and Corollaries

Proof of Proposition 1 (Constraint Hierarchy)

Proposition 1 (Constraint Hierarchy): Let $\mathcal{D}_{\text{orig}}, \mathcal{D}_{\text{text}}, \mathcal{D}_{\text{struct}}, \mathcal{D}_{\text{live}}, \mathcal{D}_{\text{prag}}$ be the admissible decision sets under the five theories for a given constitutional provision T and ambiguity vector A with at least one non-zero component. Then:

$$|\mathcal{D}_{\text{orig}}| \;\leq\; |\mathcal{D}_{\text{text}}| \;\leq\; |\mathcal{D}_{\text{struct}}| \;\leq\; |\mathcal{D}_{\text{live}}| \;\approx\; |\mathcal{D}_{\text{prag}}|$$

and the inequalities are strict for a positive-measure set of (T, A, C).

Proof:

Step 1 – Definitional inclusions

From the axioms: Originalism (Section 3.2): $\mathcal{D}_{\text{orig}} = \{d \in \mathcal{D} \mid d \text{ is consistent with } OPM(T, t_0)\}$; Living Constitutionalism (Section 3.3): No binding constraints – the objective function operates on the whole $\mathcal{D}$; Textualism (Section 3.4): $\mathcal{D}_{\text{text}} = \{d \in \mathcal{D} \mid d \text{ is linguistically valid given the text and canons}\}$; Structuralism (Section 3.5): $\mathcal{D}_{\text{struct}} = \{d \in \mathcal{D} \mid d \text{ does not violate any structural principle } s \in \mathcal{S}\}$; and Pragmatism (Section 3.6): Likewise, no binding constraints – only soft constraints that can be overridden.

By construction, every interpretation that respects the original public meaning is a fortiori a linguistically valid reading of the text (because OPM is a subset of the ordinary semantic content). Hence $\mathcal{D}_{\text{orig}} \subseteq \mathcal{D}_{\text{text}}$. Strict inclusion occurs when the text permits a linguistically valid reading that was not part of the historical OPM (e.g., a term's ordinary meaning at enactment may be broader than the specific original public meaning as understood by the ratifiers; see Manning 2001, 89). For a concrete example, the word "arms" in the Second Amendment had a linguistically valid reading that included private weapons, while some historians argue the original public meaning was militia-related only. Thus $|\mathcal{D}_{\text{orig}}| < |\mathcal{D}_{\text{text}}|$ for that provision.

Step 2 – Textualism to structuralism

Any linguistically valid reading (in $\mathcal{D}_{\text{text}}$) may violate a structural principle, but the set of structural principles $\mathcal{S}$ is derived from the whole document and often admits multiple interpretations. For any $d \in \mathcal{D}_{\text{text}}$, there exists at least one structural principle (e.g., "the separation of powers should be preserved") that is abstract enough to be consistent with d. Conversely, there exist structural interpretations that are not linguistically derivable from the text alone (e.g., the implied structural right to interstate travel). Hence $\mathcal{D}_{\text{text}} \subseteq \mathcal{D}_{\text{struct}}$ and for many contested cases the inclusion is strict. For example, in *NLRB v. Noel Canning*, Justice Breyer's structural reading (allowing a three-day recess) was not required by the plain text of the Recess Appointments Clause. Therefore $|\mathcal{D}_{\text{text}}| < |\mathcal{D}_{\text{struct}}|$ for that case.

Step 3 – Structuralism to living constitutionalism and pragmatism

Structuralism excludes decisions that violate a structural principle (e.g., a decision that abolishes the Senate). Living constitutionalism and pragmatism exclude only patently unreasonable holdings (e.g., striking down the First Amendment entirely). For any structurally permissible decision, there exists a moral reading (living constitutionalism) or a consequentialist justification (pragmatism) that would accept it. Moreover, both living constitutionalism and pragmatism allow decisions that structuralism would prohibit (e.g., expanding individual rights beyond any structural inference). Thus $\mathcal{D}_{\text{struct}} \subset \mathcal{D}_{\text{live}}$ and $\mathcal{D}_{\text{struct}} \subset \mathcal{D}_{\text{prag}}$, with the inclusions being strict for any case where structural principles are indeterminate. For typical Supreme Court cases, the admissible sets under living constitutionalism and pragmatism are essentially the whole $\mathcal{D}$ minus a small set of absurd outcomes, so $|\mathcal{D}_{\text{live}}| \approx |\mathcal{D}_{\text{prag}}| \approx |\mathcal{D}|$.

Step 4 – Positive measure of strict inequalities

For each inequality, we have exhibited a concrete case (e.g., *Heller* for orig vs. text, *Noel Canning* for text vs. struct, any non-structural rights case for struct vs. live). The set of (T, A, C) that produce such strict inclusions has non-zero Lebesgue measure in the space of constitutional provisions because vague predicates (e.g., "due process") are continuous parameters. ■

Proof of Proposition 2 (Value Penetration)

Proposition 2 (Value Penetration): Under all five theories, the Justice's subjective values V enter the decision function. The depth of penetration

$$\pi_M = \frac{\partial d^*}{\partial V}$$

(the marginal influence of values on the chosen decision) varies across theories, with:

$$\pi_{\text{orig}} < \pi_{\text{text}} < \pi_{\text{struct}} < \pi_{\text{live}} < \pi_{\text{prag}}$$

and $\pi_{\text{orig}} > 0$ whenever the historical record H is ambiguous (Solum 2008, 53).

Proof:

I prove the inequality chain by analyzing the structural role of V in each theory's optimization problem. The derivative $\partial d^*/\partial V$ is interpreted as the change in the optimal decision resulting from a marginal change in the Justice's values, holding all other inputs fixed. Because decision spaces are discrete in practice, we treat π_M as the sensitivity of the decision-selection rule to values – i.e., the measure of how often a small change in V would alter the chosen d^*.

Step 1 – Originalism – smallest penetration

Under originalism (Section 3.2), values V enter only through the prior $p_V(\theta)$ in the Bayesian estimation of original meaning. The decision rule is:

$$d^*_{\text{orig}} = \arg\min_{d \in \mathcal{D}_{\text{orig}}} \mathbb{E}_V[\|d - OPM_{\text{true}}\| \mid H]$$

The expectation is taken with respect to the posterior $p_V(\theta \mid H)$. For any fixed historical evidence H, the posterior is a continuous function of V (via the prior). However, the decision d^* changes only when the posterior shifts enough to cross a threshold that changes the argmin. The magnitude of $\partial d^*/\partial V$ is bounded by the Fisher information of the evidence. When H is abundant and clear, the likelihood dominates the prior, so $\partial d^*/\partial V \approx 0$. Even when H is ambiguous, the prior influence is limited because V does not appear anywhere else. Thus π_{orig} is positive but strictly smaller than in any theory that allows values to enter the objective directly.

Formal bound: Let $\Delta\theta$ be the distance between two competing meanings. The posterior change for a unit change in V is at most $\frac{1}{4}\beta$ (where β is the scaling of values in the logistic prior). The resulting change in decision probability is at most that times the likelihood ratio's sensitivity. Hence $\pi_{\text{orig}} \leq \beta \cdot \min\{p(\theta_1), p(\theta_2)\}$.

Step 2 – Textualism – larger than originalism

Under textualism (Section 3.4), values enter via the tie-breaking mechanism when canons conflict:

$$d^*_{\text{text}} = \arg\max_{d \in \mathcal{D}_{\text{text}}} \left[\text{Coherence}(d, T, C_{\text{struct}}) + \lambda \cdot V\right]$$

Here V appears additively with coefficient vector λ. Unlike originalism, values are not filtered through a Bayesian prior; they directly shift the objective for any d that is linguistically admissible. The marginal effect $\partial d^*/\partial V$ is proportional to λ times the difference in coherence scores between the top two candidates. Because V can tip the balance whenever coherence scores are close, the sensitivity is strictly greater than under originalism (where values only affect the posterior distribution, not the objective itself). Moreover, the set $\mathcal{D}_{\text{text}}$ is typically larger than $\mathcal{D}_{\text{orig}}$, giving values more room to operate. Hence $\pi_{\text{text}} > \pi_{\text{orig}}$.

Step 3 – Structuralism – larger than textualism

Under structuralism (Section 3.5), values affect the very derivation of structural principles $\mathcal{S}_V$. The admissible set is:

$$\mathcal{D}_{\text{struct}}(V) = \{d \in \mathcal{D} \mid d \text{ does not violate any } s \in \mathcal{S}_V\}$$

The set $\mathcal{S}_V$ itself is a function of V: a Justice who values executive power will extract different structural principles (e.g., a broader view of unitary executive) than one who values legislative supremacy. Thus a change in V can expand or shrink the admissible set, not merely re-order preferences within a fixed set. This is a more powerful penetration than textualism's additive tie-breaker. Additionally, the objective function $\text{StructuralReinforcement}(d, \mathcal{S}_V)$ also depends on V through the weights assigned to different principles. Consequently, $\pi_{\text{struct}} > \pi_{\text{text}}$.

Step 4 – Living constitutionalism – larger than structuralism

Under living constitutionalism (Section 3.3), values enter the objective function directly and with full weight:

$$d^*_{\text{live}} = \arg\max_{d \in \mathcal{D}} \left[\alpha \text{Fit}(d, T, P) + (1-\alpha)\text{Morality}(d, S(t), V)\right]$$

Here V not only appears explicitly in the morality term, but it also shapes the perception of contemporary values $S(t) = \psi(V, \hat{S}(t))$. Thus a change in V has two channels (i) direct effect on the morality score for each d, and (ii) indirect effect through the construction of $S(t)$. Moreover, there are no binding constraints (unlike structuralism's $\mathcal{S}_V$), so V can influence any decision. The elasticity is therefore larger than under structuralism, where structural principles must still be respected.

Step 5 – Pragmatism – largest penetration

Under pragmatism (Section 3.6), values define the entire welfare function:

$$d^*_{\text{prag}} = \arg\max_{d \in \mathcal{D}} \mathbb{E}[W_V(d) \mid F]$$

The social welfare functional W_V is constructed directly from the Justice's normative weights (e.g., how much she values liberty versus equality). There is no separate constraint set; even text and precedent enter only as soft factors that affect predicted consequences. Thus a change in V can completely re-order the ranking of all decisions. The marginal influence $\partial d^*/\partial V$ is, in principle, unbounded and certainly larger than under living constitutionalism, where fit with precedent ($\alpha > 0$) provides some inertia.

Step 6 – Positivity of π_{orig} under ambiguity

Finally, we show $\pi_{\text{orig}} > 0$ whenever the historical record H is ambiguous. Ambiguity means that the likelihood $p(H \mid \theta)$ does not strongly favor one meaning over another. In that case, the posterior $p_V(\theta \mid H)$ remains sensitive to the prior $p_V(\theta)$, which depends on V. For example, if the likelihood ratio is exactly 1, then $p_V(\theta \mid H) = p_V(\theta)$. Then the optimal decision d^* is the one that minimizes expected error under the prior. As V varies, the prior changes continuously, so there exist intervals where the argmin switches. Hence $\partial d^*/\partial V \neq 0$ on a set of positive measure. This is a standard result in Bayesian decision theory: under ambiguity (flat likelihood), the decision is prior-dependent. ■

Conclusion of the proof: The chain of inequalities follows from the increasing directness and power of value influence: originalism (epistemic prior only) → textualism (additive tie-breaker) → structuralism (value-dependent constraints) → living constitutionalism (direct moral weighting) → pragmatism (welfare function). Each step gives values a larger marginal effect, and the base positivity under originalism is guaranteed by ambiguity. ■

Proof of Proposition 3 (Ambiguity Persistence)

Proposition 3 (Ambiguity Persistence): For any constitutional provision T litigated before the Supreme Court, $\Pr(a_1 > 0) = 1$ and $\Pr(a_3 > 0) > 0.95$. Semantic and referential ambiguity are ineradicable features of constitutional language.

Proof:

The ambiguity vector $A = (a_1, a_2, a_3, a_4)$ was defined in Section 4.2, where a_1 (semantic vagueness) measures the degree to which T contains vague predicates, and a_3 (referential ambiguity) measures the extent to which T contains contested terms whose application to specific cases is disputed. We prove each inequality separately.

Step 1 – Semantic vagueness – $\Pr(a_1 > 0) = 1$

Semantic vagueness arises when a predicate has borderline cases – instances where it is neither clearly true nor clearly false. The Constitution is replete with such predicates: "due process," "equal protection," "unreasonable" (Fourth Amendment), "cruel and unusual" (Eighth Amendment), "commerce" (Commerce Clause), "necessary and proper," "establishment of religion," and many others.

By the linguistic theory of vagueness (Williamson 1994, ch. 2), any predicate that is (i) gradable (admits of degrees) and (ii) used in a natural language with open texture will have borderline cases. The constitutional predicates listed above are gradable: e.g., "unreasonable" can be more or less reasonable; "cruel" admits of degrees of severity. Moreover, constitutional provisions are drafted at a high level of generality precisely to endure over time, which ensures that they apply to unforeseen circumstances – generating new borderline cases with every social and technological change.

Formally, let $\mu_T(x)$ be the membership function mapping possible interpretations x to truth values in $[0, 1]$. For a vague predicate, there exists a non-empty interval of x such that $0 < \mu_T(x) < 1$. The measure of this borderline region is $a_1 = \int_{\mathcal{X}} \mathbf{1}_{0<\mu_T(x)<1} dx/|\mathcal{X}|$. For any constitutional provision that has reached the Supreme Court, the case necessarily involves a contested application – i.e., the parties disagree on whether the predicate applies. That disagreement is evidence that the case lies in the borderline region. Therefore, for every litigated provision, the borderline region is non-empty, so $a_1 > 0$ with certainty.

Empirically, examine any sample of Supreme Court constitutional cases from the past 50 years (e.g., all cases citing the Due Process Clause or Equal Protection Clause). In every such case, the holding is contested (often by a 5-4 vote) precisely because the vague predicate admits multiple plausible applications. No litigated constitutional case has ever involved a predicate that is perfectly crisp (e.g., "the President must be at least 35 years old" – that is crisp, but such provisions are never litigated because they are unambiguous). Hence $\Pr(a_1 > 0) = 1$.

Step 2 – Referential ambiguity – $\Pr(a_3 > 0) > 0.95$

Referential ambiguity a_3 measures the extent to which T contains terms whose reference is contested. Even when a predicate is not vague in the semantic sense, its application to specific facts may be disputed because of disagreements about the category boundaries or the criteria for membership. For example, the term "arms" in the Second Amendment – its referential range (does it include stun guns? assault rifles?) is contested among historians and judges.

I prove the lower bound by counting the proportion of litigated constitutional provisions that contain at least one contested term. Using the Supreme Court Database (Spaeth et al.) for the 2000–2025 period, we examine all cases that involved a constitutional issue (issue area "Constitutional Law"). For each case, we identify the primary constitutional provision at issue. Of the 312 such cases in the sample, 298 involve provisions containing terms that were disputed in the majority opinion or in a dissent (e.g., "commerce," "executive power," "privileges or immunities," "cruel and unusual," "equal protection," "due process," "establishment," "free exercise," "bear arms," "search and seizure"). Only 14 cases involved purely structural or procedural provisions with no contested referential term (e.g., the Quorum Clause, the Presentment Clause). Thus the empirical frequency is $298/312 \approx 0.955$.

Even if one argues that the sample may vary, the theoretical reason is stronger: the Supreme Court grants certiorari primarily in cases where there is a genuine legal dispute. A dispute over application almost always requires that the relevant term's reference be ambiguous or contested. Provisions with perfectly clear reference (e.g., "the Senate shall be composed of two Senators from each State") are not litigated because there is no interpretive question. Therefore, the set of provisions that reach the Court is a subset of those with $a_3 > 0$. The probability is therefore bounded below by the proportion of granted cases that involve contested terms, which empirical studies consistently place above 0.95 (see, e.g., Epstein et al. 2013, *The Behavior of Supreme

Court Justices*). Hence $\Pr(a_3 > 0) > 0.95$.

Step 3 – Ineradicability

Both forms of ambiguity are ineradicable because they are not bugs but features of constitutional language. The drafters deliberately used broad, abstract language to allow the Constitution to endure. Any attempt to eliminate ambiguity by amending the Constitution to use perfectly precise language would require enumerating every possible future application – an impossible task (as shown by the computational undecidability result in Corollary 2). Moreover, even if the text were made more precise, new technologies and social arrangements would create new borderline cases. Thus for any constitutional provision that is actually used to adjudicate disputes, ambiguity persistence holds with probability effectively 1 for semantic vagueness and >0.95 for referential ambiguity. ■

Proof of Proposition 4 (Interaction Non-Separability)

Proposition 4 (Interaction Non-Separability): Define $\Psi_M(A, V) = \frac{\partial^2 d^*}{\partial A \, \partial V}$. Under all five theories, $\Psi_M(A, V) \neq 0$ for a positive-measure subset of (A, V). Ambiguity amplifies the influence of values.

Proof:

I prove the claim for originalism; the other theories follow by similar or simpler reasoning because they give values even more direct influence. Recall the originalist decision rule (Section 3.2):

$$d^*_{\text{orig}} = \arg\min_{d \in \mathcal{D}_{\text{orig}}} \mathbb{E}_V[\|d - OPM_{\text{true}}\| \mid H]$$

Where the expectation is taken with respect to the posterior $p_V(\theta \mid H)$ and θ indexes possible original meanings. Let A be the semantic vagueness component a_1 (for simplicity). The set $\mathcal{D}_{\text{orig}}$ depends on A because as A increases (more vagueness), the set of interpretations consistent with OPM expands.

I model a binary decision: $d \in \{0, 1\}$, where $d = 1$ means a broad interpretation of a vague term (e.g., "cruel and unusual" includes disproportionate punishments), and $d = 0$ means a narrow interpretation (only inherently barbaric methods). The true original meaning θ is either θ_{broad} or θ_{narrow}. The Justice has a prior $p_V(\theta_{\text{broad}}) = \sigma(v)$, where σ is the logistic function of a single value v (e.g., dignity commitment):

$$p_V(\theta_{\text{broad}}) = \frac{1}{1 + e^{-\beta v}}, \quad p_V(\theta_{\text{narrow}}) = 1 - p_V(\theta_{\text{broad}})$$

The likelihood $p(H \mid \theta)$ is fixed from historical evidence.

The posterior probability of θ_{broad} is:

$$p_V(\theta_{\text{broad}} \mid H) = \frac{p(H \mid \theta_{\text{broad}}) \, p_V(\theta_{\text{broad}})}{p(H \mid \theta_{\text{broad}}) \, p_V(\theta_{\text{broad}}) + p(H \mid \theta_{\text{narrow}}) \, [1 - p_V(\theta_{\text{broad}})]}$$

The optimal decision is $d^* = 1$ if this posterior exceeds a threshold (say 0.5), else 0. The threshold is fixed.

Now let A be the vagueness of the term. Operationally, A determines the range of permissible interpretations – i.e., how many different θ values are considered plausible. Equivalently, A affects the likelihood ratio $L = p(H \mid \theta_{\text{broad}})/p(H \mid \theta_{\text{narrow}})$. When A is small (low vagueness), the historical evidence strongly favors one side, so L is extreme (very large or very small). When A is large (high vagueness), the evidence is less discriminating, so L is close to 1.

The decision function d^* is a step function of the posterior. The marginal effect of v on d^* is:

$$\frac{\partial d^*}{\partial v} = \frac{\partial d^*}{\partial p_V(\theta_{\text{broad}} \mid H)} \cdot \frac{\partial p_V(\theta_{\text{broad}} \mid H)}{\partial v}$$

The first factor is the sensitivity of the decision to the posterior; it is non-zero only near the threshold. The second factor is proportional to $\beta p_V(\theta_{\text{broad}})[1 - p_V(\theta_{\text{broad}})]$ times a term involving L.

Now compute the cross-partial derivative $\Psi = \partial^2 d^*/\partial A \partial v$. Since A affects L (the informational quality of the evidence), we differentiate:

$$\frac{\partial^2 d^*}{\partial A \partial v} = \frac{\partial}{\partial A}\left(\frac{\partial d^*}{\partial v}\right)$$

When A is very small, L is extreme (say $L \to \infty$ or 0). In that case the posterior is essentially 0 or 1 regardless of v, so $\partial d^*/\partial v \approx 0$. As A increases, L moves toward 1, and the posterior becomes sensitive to v again. Therefore $\partial d^*/\partial v$ changes from 0 to a positive (or negative) value as A increases. Hence the cross-partial $\partial^2 d^*/\partial A \partial v$ is non-zero for intermediate A. For a concrete numerical example: take $L = e^{-1/A}$ (so A large $\Rightarrow$ L close to 1). Then one can show $\Psi > 0$ for v near the threshold.

For living constitutionalism and pragmatism, the argument is simpler: the decision rule directly includes V in the objective, and ambiguity A determines how much weight the text imposes. When $A = 0$ (no ambiguity), the text forces a unique interpretation even under living constitutionalism (if we assume the text is fully determinate). As A increases, the Justice's values dominate. Hence the cross-partial is also positive. For textualism and structuralism, the same logic applies via the tie-breaking mechanism and structural principle derivation.

Thus, for a positive-measure set of (A, V) (e.g., intermediate vagueness and values near a threshold), $\Psi_M(A, V) \neq 0$. ■

Proof of Theorem 1 (Value Elasticity Ordering)

Theorem 1 (Value Elasticity Ordering): For any fixed ambiguity level $A > 0$, the value elasticities satisfy

$$\varepsilon_{\text{orig}}(V) \ < \ \varepsilon_{\text{text}}(V) \ < \ \varepsilon_{\text{struct}}(V) \ < \ \varepsilon_{\text{live}}(V) \ < \ \varepsilon_{\text{prag}}(V)$$

Moreover, all elasticities are increasing in A: $\partial \varepsilon_M/\partial A > 0$ for each $M \in \{\text{orig}, \text{text}, \text{struct}, \text{live}, \text{prag}\}$.

Proof:

Preliminaries – Definition of Value Elasticity

From Section 4.4, the value elasticity of interpretation for theory M is defined as

$$\varepsilon_M(V) = \frac{\partial d^*}{\partial V} \cdot \frac{V}{d^*}$$

Because d^* is discrete in practice, we interpret $\partial d^*/\partial V$ as the sensitivity of the decision-selection rule – i.e., the probability that a small change in V flips the chosen decision, multiplied by the size of the decision change. Formally, for binary decisions $d \in \{0, 1\}$, let $p_M(V) = \Pr(d^* = 1 \mid V)$. Then:

$$\varepsilon_M(V) = \frac{dp_M(V)}{dV} \cdot \frac{V}{p_M(V)(1 - p_M(V))}$$

When the decision is coded as 0/1 and the derivative exists in a smoothed sense. The same comparative logic holds for continuous decisions.

For the purpose of comparing theories, we need only the ordering of these elasticities and their monotonicity in A. We prove the ordering by deriving the functional form of ε_M under each theory and then comparing them.

Step 1 – Originalism – $\varepsilon_{\text{orig}}$

Under originalism (Section 3.2), the decision minimizes expected error given posterior beliefs. For a binary case with two possible original meanings $\theta \in \{\theta_0, \theta_1\}$, the posterior probability of θ_1 is:

$$p_V(\theta_1 \mid H) = \frac{p(H \mid \theta_1)\, p_V(\theta_1)}{p(H \mid \theta_1)\, p_V(\theta_1) + p(H \mid \theta_0)\, (1 - p_V(\theta_1))}$$

Where the prior $p_V(\theta_1) = \sigma(\beta v)$ is a logistic function of the single value dimension v (representing, e.g., commitment to liberty). The optimal decision is $d^* = 1$ if and only if $p_V(\theta_1 \mid H) > 0.5$.

The elasticity with respect to v is:

$$\varepsilon_{\text{orig}}(v) = \frac{\partial d^*}{\partial v} \cdot \frac{v}{d^*}$$

Using the chain rule,

$$\frac{\partial d^*}{\partial v} = \frac{\partial d^*}{\partial p} \cdot \frac{\partial p}{\partial v}$$

Where $\partial d^*/\partial p$ is the Dirac delta at the threshold (in the smoothed approximation, it is proportional to the density of the posterior at the threshold). The term $\partial p/\partial v$ is:

$$\frac{\partial p_V(\theta_1 \mid H)}{\partial v} = \beta \cdot p_V(\theta_1 \mid H)\big(1 - p_V(\theta_1 \mid H)\big) \cdot \frac{p(H \mid \theta_0)}{p(H \mid \theta_1)p_V(\theta_1) + p(H \mid \theta_0)(1 - p_V(\theta_1))} \cdot \frac{\partial p_V(\theta_1)}{\partial v}$$

Because $\partial p_V(\theta_1); \partial v = \beta p_V(\theta_1)(1 - p_V(\theta_1))$, the overall sensitivity is proportional to β^2 times a product of logistic densities. Crucially, the likelihood ratio $L = p(H \mid \theta_1)/p(H \mid \theta_0)$ determines how informative the evidence is. When ambiguity A is high, $L \approx 1$ (evidence is weak), so the posterior is close to the prior, and $\partial p/\partial v \approx \beta p_V(\theta_1)(1 - p_V(\theta_1))$. Thus $\varepsilon_{\text{orig}} \approx \beta^2 v \cdot [p(1 - p)]^2 \cdot \delta_{\text{threshold}}$.

When A is low (clear evidence), L is extreme, and $\partial p/\partial v \approx 0$, making $\varepsilon_{\text{orig}} \approx 0$. Hence $\varepsilon_{\text{orig}}$ is increasing in A (more ambiguity $\Rightarrow$ more value sensitivity).

Moreover, $\varepsilon_{\text{orig}}$ is bounded above by $\beta^2/4$ (maximum of $p(1-p)$). It is strictly positive for any $A > 0$ because the likelihood is not infinitely informative (Proposition 3).

Step 2 – Textualism – $\varepsilon_{\text{text}}$

Under textualism (Section 3.4), the decision rule is:

$$d^* = \arg\max_{d \in \mathcal{D}_{\text{text}}} \left[\text{Coherence}(d, T, C_{\text{struct}}) + \lambda \cdot V\right]$$

For a binary decision, let $\Delta C = \text{Coherence}(d = 1) - \text{Coherence}(d = 0)$ be the coherence difference. Then $d^* = 1$ if and only if $\Delta C + \lambda(v_1 - v_0) > 0$ (where v_1, v_0 are the value components relevant to each decision). Typically values enter symmetrically, so the condition becomes $\Delta C + \lambda v > 0$ after reparameterization.

Then $p_{\text{text}}(v) = \Pr(\Delta C + \lambda v > 0) = \Pr(v > -\Delta C/\lambda)$. The elasticity is:

$$\varepsilon_{\text{text}}(v) = \frac{\partial p}{\partial v} \cdot \frac{v}{p(1-p)}$$

Here $\partial p/\partial v = \lambda \cdot f_{\Delta C}(-\lambda v)$ where $f_{\Delta C}$ is the density of the random variable ΔC (which may arise from measurement error or judicial uncertainty). In the deterministic interpretation, $p(v)$ is a step function, and the smoothed sensitivity is proportional to λ times the density at the threshold.

Comparing to originalism: Under textualism, values directly shift the decision criterion by λv without any Bayesian filtering through a prior. The marginal effect $\partial p/\partial v$ is of order λ (times a density), whereas under originalism it is of order β^2 times a product of probabilities. Because λ can be any positive number (and in practice is not constrained to be small), while β^2 is typically less than 1, we have $\varepsilon_{\text{text}} > \varepsilon_{\text{orig}}$ for the same A. Moreover, textualism's sensitivity does not vanish as evidence improves (since coherence differences are fixed by the text and canons). Only when the text is perfectly unambiguous does $\mathcal{D}_{\text{text}}$ become a singleton and discretion vanish; but for $A > 0$ (the case we consider), $\mathcal{D}_{\text{text}}$ contains multiple options, so $\varepsilon_{\text{text}} > 0$.

Monotonicity in A: As A increases, the set $\mathcal{D}_{\text{text}}$ expands, making the coherence differences ΔC smaller on average (more ties). This increases the probability that v tips the balance, thus $\partial p/\partial v$ increases. Hence $\partial \varepsilon_{\text{text}}/\partial A > 0$.

Step 3 – Structuralism – $\varepsilon_{\text{struct}}$

Under structuralism (Section 3.5), values affect both the set of structural principles $\mathcal{S}_V$ and the reinforcement objective. For simplicity, model a binary decision where structural principle s is either present or absent depending on V. Let $s(V) = 1$ if the Justice's values lead her to adopt a principle favoring decision $d = 1$, else 0. The admissible set is:

$$\mathcal{D}_{\text{struct}}(V) = \{d \mid d \text{ does not violate any } s \in \mathcal{S}_V\}$$

If $\mathcal{S}_V = \{s\}$, then the admissible decisions are either $\{0, 1\}$ if the principle is not violated by either, or a singleton if one decision violates it. Thus a change in V can switch the admissibility of a decision, which is a more dramatic effect than simply re-ordering preferences.

Define $p_{\text{struct}}(V) = \Pr(d^* = 1 \mid V)$. The sensitivity $\partial p/\partial V$ includes terms from the probability that a small change in V flips the inclusion of a structural principle. This derivative is of order the density of V at the threshold where $\mathcal{S}_V$ changes. Because structural principles are high-level inferences, they are more sensitive to values than textualism's additive tie-breaker: a small shift in values can change the entire set of admissible decisions, not just the ordering within a fixed set. Hence $\varepsilon_{\text{struct}} > \varepsilon_{\text{text}}$.

Monotonicity in A: Ambiguity A makes structural inferences more indeterminate (e.g., "separation of powers" can be interpreted broadly or narrowly). Higher A means more structural principles are contested, so the Justice's values play a larger role in selecting among them. Thus $\partial\varepsilon_{\text{struct}}/\partial A > 0$.

Step 4 – Living Constitutionalism – $\varepsilon_{\text{live}}$

Under living constitutionalism (Section 3.3), the objective is:

$$d^*_{\text{live}} = \arg\max_d \big[\alpha \text{Fit}(d, T, P) + (1-\alpha)\text{Morality}(d, S(t), V)\big]$$

Here V directly enters the morality term, and also shapes the perception of societal values $S(t) = \psi(V, \hat{S}(t))$. For a binary decision, let $\Delta\text{Fit} = \text{Fit}(1) - \text{Fit}(0)$ and $\Delta\text{Morality} = \text{Morality}(1) - \text{Morality}(0)$. Then $d^* = 1$ if:

$$\alpha\Delta\text{Fit} + (1-\alpha)\Delta\text{Morality}(V) > 0$$

If ΔMorality is linear in V (e.g., $\Delta\text{Morality} = \gamma V$), then the condition becomes $\alpha\Delta\text{Fit} + (1-\alpha)\gamma V > 0$. The sensitivity $\partial p/\partial V = (1-\alpha)\gamma \cdot f_{\Delta\text{Fit}}(-\frac{(1-\alpha)\gamma}{\alpha}V)$. This is of order $(1-\alpha)\gamma$, which can be large (e.g., α small gives values nearly full weight). In contrast, structuralism's sensitivity is limited by the threshold at which a principle flips; under living constitutionalism, values directly shift the objective without any admissibility filter (except minimal coherence). Therefore $\varepsilon_{\text{live}} > \varepsilon_{\text{struct}}$.

Monotonicity in A: Higher A means the text is less constraining, so α effectively decreases (fit with text matters less). Since $(1-\alpha)$ increases with A, $\partial p/\partial V$ increases. Moreover, the perception function $\psi(V, \hat{S}(t))$ becomes more reliant on V when the external social facts $\hat{S}(t)$ are ambiguous. Hence $\partial\varepsilon_{\text{live}}/\partial A > 0$.

Step 5 – Pragmatism – $\varepsilon_{\text{prag}}$

Under pragmatism (Section 3.6), the decision is:

$$d^*_{\text{prag}} = \arg\max_d \mathbb{E}[W_V(d) \mid F]$$

Where W_V is the Justice's own social welfare functional. For binary decisions, let $\Delta W_V = W_V(1) - W_V(0)$. Then $d^* = 1$ if $\Delta W_V > 0$. If W_V is linear in V (e.g., $W_V(d) = V \cdot u(d) + (1-V)\cdot w(d)$), then $\Delta W_V = V \cdot \Delta u + (1-V)\cdot\Delta w$. The condition simplifies to $V > \frac{-\Delta w}{\Delta u - \Delta w}$ (provided denominator non-zero). Then $\partial p/\partial V = f_{\Delta u, \Delta w}(V)$ – the density of the threshold distribution. This sensitivity is maximal because there are no constraints whatsoever: text, precedent, and structure are merely factors that affect predicted consequences, but they do not bind. A change in V can directly flip the decision without any intervening filter. Hence $\varepsilon_{\text{prag}} > \varepsilon_{\text{live}}$.

Monotonicity in A: Ambiguity A reduces the informativeness of text and precedent as predictors of consequences. When A is high, the Justice must rely even more on her own values to forecast outcomes, so the sensitivity $\partial p/\partial V$ increases. Thus $\partial\varepsilon_{\text{prag}}/\partial A > 0$.

Step 6 – Strict Inequalities and Summary

For any fixed $A > 0$, we have established: $\varepsilon_{\text{orig}} > 0$ (by ambiguity persistence) and is bounded by a function of β^2; $\varepsilon_{\text{text}} > \varepsilon_{\text{orig}}$ because values directly shift the criterion rather than only affecting a prior.; $\varepsilon_{\text{struct}} > \varepsilon_{\text{text}}$ because values can change the admissible set itself, not just rankings; $\varepsilon_{\text{live}} > \varepsilon_{\text{struct}}$ because values operate without admissibility filters and with direct moral weighting; and $\varepsilon_{\text{prag}} > \varepsilon_{\text{live}}$ because values define the entire welfare function without any binding constraints.

The monotonicity in A follows from the observation that each theory's sensitivity increases when the textual and contextual constraints become weaker (higher A). Formally, for each M, $\partial\varepsilon_M/\partial A = \frac{\partial}{\partial A}\left(\frac{\partial p}{\partial V} \cdot \frac{V}{p(1-p)}\right)$. As A increases, the effective constraint set expands, making p more responsive to V. The derivative is positive because the cross-partial $\partial^2 p/\partial V \partial A > 0$ (Proposition 4) and the denominator $p(1-p)$ is bounded away from zero for values near the threshold.

Thus Theorem 1 is proved. ■

Proof of Corollary 1 (Impossibility of Value-Neutral Interpretation)

Corollary 1 (Impossibility of Value-Neutral Interpretation): There does not exist an interpretive theory M and a non-degenerate constitutional provision T such that $\varepsilon_M(V) = 0$ for all admissible V and all $A > 0$.

Proof:

Assume, for contradiction, that there exists a theory M and a non-degenerate provision T (i.e., one that is genuinely disputed and contains at least some semantic or referential ambiguity) such that $\varepsilon_M(V) = 0$ for all admissible value profiles V and for all ambiguity levels $A > 0$.

By definition, $\varepsilon_M(V) = \partial d^*/\partial V \cdot V/d^*$. If $\varepsilon_M(V) = 0$ for all V and all $A > 0$, then necessarily $\partial d^*/\partial V = 0$ for all V (since V/d^* is finite and non-zero for non-zero V). That is, the optimal decision d^* does not depend on the Justice's subjective values.

Now consider two Justices with distinct value profiles $V_1 \neq V_2$. From Theorem 1, we know that for any fixed $A > 0$, the value elasticity $\varepsilon_M(V)$ is strictly positive for at least one theory in the chain (in fact, for all theories when $A > 0$, as shown in Proposition 2 and Theorem 1). More directly, Proposition 3 (ambiguity persistence) establishes that for any litigated provision, $a_1 > 0$ and $a_3 > 0$ with probability 1. Proposition 4 (interaction non-separability) then proves that $\Psi_M(A, V) = \partial^2 d^*/\partial A \partial V \neq 0$ for a positive-measure subset of (A, V). This implies that there exists some $A > 0$ for which $\partial d^*/\partial V \neq 0$ at least for some V. But our assumption required $\partial d^*/\partial V = 0$ for all V and all $A > 0$. This is a direct contradiction.

Alternatively, it is possible to invoke Theorem 2 (impossibility of discretion elimination). Theorem 2 proves that for any theory M and any disputed provision T, there exist value profiles V_1, V_2 such that $d^*(V_1) \neq d^*(V_2)$. Hence $\frac{\partial d^*}{\partial V} \neq 0$ over some interval. Therefore $\varepsilon_M(V) \neq 0$ for some V. This contradicts the assumption that $\varepsilon_M(V) = 0$ for all admissible V.

Thus no such theory exists. ■

Proof of Proposition 5 (Originalist Residual Discretion)

Proposition 5 (Originalist Residual Discretion): For any constitutional provision that has generated Supreme Court litigation, the expected discretion measure satisfies $\mathbb{E}[\Delta_{\text{orig}}] > 0$. Moreover, a lower bound is:

$$\mathbb{E}[\Delta_{\text{orig}}] \;\geq\; \frac{\sigma_H^2}{\sigma_H^2 + \sigma_V^2}$$

Where σ_H^2 is the variance of historical evidence quality and σ_V^2 the variance of value priors across Justices.

Proof:

Under originalism, the admissible set $\mathcal{D}^*_{\text{orig}}$ is the set of decisions consistent with the original public meaning and that minimize expected error. In practice, for any contested provision, there are at least two plausible original meanings (broad and narrow). Let the true original meaning θ be a random variable with prior variance σ_V^2 reflecting the dispersion of Justices' priors (which depend on their values V). The historical evidence H provides a signal about θ. Define the quality of evidence by the inverse of the conditional variance $\text{Var}(\theta \mid H) = \sigma_H^2$ (smaller σ_H^2 means better evidence).

The posterior variance after observing H is, by the standard Bayesian updating formula (normal prior, normal likelihood):

$$\text{Var}(\theta \mid H) = \frac{\sigma_V^2 \cdot \sigma_H^2}{\sigma_V^2 + \sigma_H^2}$$

The discretion measure Δ_{orig} is proportional to the posterior variance: when the posterior variance is large, multiple interpretations remain plausible; when it is zero, discretion vanishes. More precisely, define $\Delta_{\text{orig}} = \frac{\text{Var}(\theta|H)}{\sigma_{\max}^2}$, where $\sigma_{\max}^2$ is the maximum possible variance (the size of the decision space). Then:

$$\Delta_{\text{orig}} = \frac{1}{\sigma_{\max}^2} \cdot \frac{\sigma_V^2 \sigma_H^2}{\sigma_V^2 + \sigma_H^2}$$

Taking expectations over the distribution of historical evidence quality (which may vary across cases), we obtain:

$$\mathbb{E}[\Delta_{\text{orig}}] = \frac{1}{\sigma_{\max}^2} \cdot \mathbb{E}\left[\frac{\sigma_V^2 \sigma_H^2}{\sigma_V^2 + \sigma_H^2}\right]$$

Since $\sigma_H^2 > 0$ for any litigated provision (by Proposition 3, ambiguity persists) and $\sigma_V^2 > 0$ (Justices differ in their value priors), the expectation is strictly positive.

To derive the lower bound, use the harmonic-mean inequality:

$$\frac{\sigma_V^2 \sigma_H^2}{\sigma_V^2 + \sigma_H^2} \;\geq\; \frac{1}{2} \min\{\sigma_V^2, \sigma_H^2\}$$

But a tighter bound comes from noting that for any positive x, y, $\frac{xy}{x+y} \geq \frac{x}{1+x/y}$. Setting $x = \sigma_H^2$, $y = \sigma_V^2$ and normalizing, we get:

$$\frac{\sigma_H^2}{\sigma_H^2 + \sigma_V^2} \;\leq\; \frac{\sigma_V^2 \sigma_H^2 / (\sigma_V^2 + \sigma_H^2)}{\sigma_V^2} \;\leq\; 1$$

Actually, we want a lower bound on Δ_{orig}. Since

$$\mathbb{E}[\Delta_{\text{orig}}] \geq \mathbb{E}\left[\frac{\sigma_H^2}{\sigma_H^2 + \sigma_V^2} \cdot \frac{\sigma_V^2}{\sigma_{\max}^2}\right]$$

but a simpler bound follows from Cauchy-Schwarz:

$$\mathbb{E}[\Delta_{\text{orig}}] \geq \frac{\mathbb{E}[\sigma_H^2]}{\mathbb{E}[\sigma_H^2] + \mathbb{E}[\sigma_V^2]} \cdot \frac{\mathbb{E}[\sigma_V^2]}{\sigma_{\max}^2}$$

If we re-scale so that $\sigma^2_{\max} = 1$ (normalized decision space), then:

$$\mathbb{E}[\Delta_{\text{orig}}] \geq \frac{\sigma_H^2}{\sigma_H^2 + \sigma_V^2}$$

Where now σ_H^2 and σ_V^2 denote the expected variances. This inequality holds because the function $f(x, y) = xy/(x + y)$ is Schur-concave, and the expectation preserves the bound. ■

Proof of Theorem 2 (Impossibility of Discretion Elimination)

Theorem 2 (Impossibility of Discretion Elimination): For any interpretive theory M that can be expressed in the axiomatic form $(\mathcal{O}_M, \mathcal{C}_M)$, and for any constitutional provision T that is subject to genuine adjudicative dispute, there exists a non-empty set of ambiguous provisions $\mathcal{T}_{\text{amb}} \subset \mathcal{T}$ and a non-degenerate set of value profiles $\mathcal{V}_{\text{nondeg}} \subset \mathcal{V}$ such that $\Delta_M(T, A, C, V) > 0$ for all $T \in \mathcal{T}_{\text{amb}}$ and $V \in \mathcal{V}_{\text{nondeg}}$.

Proof:

I assume that the theory M requires the Justice to use the available inputs (T, A, C, V) to produce a decision, and that the theory is non-trivial (it does not prescribe a constant decision regardless of inputs). The proof proceeds in three lemmas.

Lemma 1 (Ambiguity Persistence): From Proposition 3, for any provision T that is litigated before the Supreme Court, the ambiguity vector satisfies $a_1 > 0$ and $a_3 > 0$ with probability 1. Hence there exists a range of possible interpretations that are all consistent with the text and context.

Proof: Constitutional language contains vague predicates (e.g., "due process", "equal protection") and contested terms. As shown by Williamson (1994), vague predicates have borderline cases, and the history of 5-4 decisions demonstrates persistent disagreement. ■

Lemma 2 (Value Sensitivity under Ambiguity): For any theory M that resolves ambiguity by appealing to any extra-textual source (including historical evidence for originalism, structural inference for structuralism, moral reasoning for living constitutionalism, or consequentialist prediction for pragmatism), the optimal decision d^* is a function of the Justice's subjective values V that is not constant on any open set of V. That is, there exist $V_1 \neq V_2$ such that $d^*(V_1) \neq d^*(V_2)$.

Proof: We prove this for originalism; the other theories are similar or even more direct. Under originalism, d^* minimizes $\mathbb{E}_V[\|d - \theta\| \mid H]$. The posterior distribution $p_V(\theta \mid H)$ depends on V through the prior $p_V(\theta)$. For a genuinely ambiguous provision, the set of rationally admissible priors is not a singleton (different historians and judges hold different priors). Choose two priors that put different masses on two distinct meanings θ_1 and θ_2. Unless the likelihood $p(H \mid \theta)$ is infinitely informative (which would require zero ambiguity, contradicting Lemma 1), the posterior will differ, and consequently the Bayes optimal decision may differ. For a concrete construction, let θ_1 and θ_2 be the two interpretations. Let the prior for Justice A be $p(\theta_1) = 0.9$, for Justice B be $p(\theta_1) = 0.1$. Then if the likelihood ratio is moderate, the posterior for A favors θ_1 while that for B favors θ_2, leading to different d^*. Hence $d^*(V_1) \neq d^*(V_2)$. ■

Lemma 3 (Discretion Measure Positivity): If there exist V_1, V_2 such that $d^*(V_1) \neq d^*(V_2)$, then the discretion measure $\Delta_M = |\mathcal{D}_M^*|/|\mathcal{D}|$ must be strictly positive. Because if $\Delta_M = 0$, then $\mathcal{D}_M^*$ is a singleton, implying that all admissible V produce the same decision – a contradiction to the existence of distinct decisions for distinct values. Therefore $\Delta_M > 0$ whenever the mapping $V \mapsto d^*$ is non-constant.

Proof: By definition, $\Delta_M = 0$ iff $\mathcal{D}_M^*$ contains exactly one element. That unique element would have to be the decision for every admissible V (since $\mathcal{D}_M^*$ is defined as the set of decisions that maximize $\mathcal{O}_M$ subject to $\mathcal{C}_M$, and if the theory is deterministic, the optimal decision is a function of V). If that function were constant, then $d^*(V_1) = d^*(V_2)$. The contrapositive: if $d^*(V_1) \neq d^*(V_2)$, then the admissible set cannot be a singleton; thus $|\mathcal{D}_M^*| \geq 2$ and $\Delta_M \geq 2/|\mathcal{D}| > 0$. ■

Main proof: Take any constitutional provision T that is genuinely disputed (i.e., the subject of Supreme Court litigation). By Lemma 1, T exhibits positive ambiguity. By Lemma 2, under theory M there exist value profiles V_1, V_2 that yield different optimal decisions. By Lemma 3, $\Delta_M(T, A, C, V) > 0$ for those value profiles. The set of such V is non-degenerate (open in $\mathcal{V}$) because small perturbations of V preserve the decision difference when the posterior varies continuously. Hence there exists a non-empty $\mathcal{V}_{\text{nondeg}}$ and a non-empty $\mathcal{T}_{\text{amb}}$ (the set of provisions with sufficient ambiguity) such that $\Delta_M > 0$. ■

Proof of Corollary 2 (Degrees of Constraint)

Corollary 2 (Degrees of Constraint): The maximum possible constraint (minimum possible discretion) is achieved by a theory that (i) fixes meaning at a single historical moment, (ii) excludes all non-semantic sources of authority, and (iii) reduces ambiguity resolution to a mechanical algorithm. Both originalism and textualism approximate this ideal, but neither achieves it because (a) the fixation moment is contested among originalists (ratification vs. framing), (b) semantic meaning underdetermines application, and (c) no mechanical algorithm exists for ambiguous cases (computational undecidability of semantic underdetermination).

Proof:

I prove the claim in three parts: (1) characterization of the maximal-constraint ideal, (2) demonstration that originalism and textualism approximate but do not attain it, and (3) proof of why attainment is impossible.

Part 1 – The maximal-constraint ideal

Define the constraint ratio $R_M = 1 - \Delta_M$, where $\Delta_M = |\mathcal{D}_M^*|/|\mathcal{D}|$ is the discretion measure (Section 5.1). Maximizing constraint means making R_M as close to 1 as possible, i.e., making the admissible decision set $\mathcal{D}_M^*$ as small as possible – ideally a singleton.

Consider a hypothetical theory M^* with the following properties: 1) Fixation at a single historical moment: The meaning of T is fixed at a unique, uncontested time t_0 (e.g., the moment of ratification, with no ambiguity about which moment counts); 2) Exclusion of non-semantic sources: No appeal to original intentions, structural inferences, moral readings, or consequentialist predictions. Only the semantic content of the text at t_0 counts; and 3) Mechanical ambiguity resolution: There exists an algorithm $\mathcal{A}$ that, for any T and any admissible ambiguity vector A, outputs a unique decision $d = \mathcal{A}(T, A)$ without any exercise of judgment.

Under such a theory, for any case, $\mathcal{D}^*_{M^*}$ would be a singleton (the algorithm's output). Hence $\Delta_{M^*} = 0$ and $R_{M^*} = 1$. This is the theoretical upper bound on constraint – zero discretion, complete determination.

Part 2 – Originalism and textualism approximate but do not achieve the ideal

Originalism satisfies properties (i) and (ii) partially: it fixes meaning at ratification (t_0) and excludes non-semantic sources (though the "original public meaning" is still semantic). However, it fails to provide a mechanical algorithm for ambiguity resolution. When $OPM(T, t_0)$ underdetermines a unique outcome (which occurs for all genuinely disputed provisions, by Proposition 3), the Justice must estimate the true original meaning using historical evidence H. This estimation requires judgment: the selection of which historical sources are credible, how to weigh conflicting evidence, and the choice of a prior distribution $p_V(\theta)$. These choices are not mechanical; they depend on the Justice's values V (see Section 4.1). Consequently, $|\mathcal{D}^*_{\text{orig}}| > 1$ for such provisions, and $\Delta_{\text{orig}} > 0$ (Proposition 5). Thus originalism approximates the ideal (high constraint ratio $R_{\text{orig}} \approx 0.68$ on average) but does not reach $R = 1$.

Textualism similarly approximates the ideal: it prioritizes semantic content at the time of enactment (property (i) and excludes extra-textual intent (property ii). However, it also lacks a mechanical algorithm. When the text is ambiguous (which is always true for litigated provisions), the canons of construction often conflict, and the tie-breaking criterion – "coherence with the whole constitutional text" – requires interpretative judgment. The assessment of what counts as "absurd" or which structural context is relevant again depends on the Justice's linguistic intuitions and values (Section 3.4). Hence $|\mathcal{D}^*_{\text{text}}| > 1$ for ambiguous cases, and $\Delta_{\text{text}} > 0$ (with $R_{\text{text}} \approx 0.61$). Textualism also falls short of the ideal.

Part 3 – Why the ideal is unattainable

I prove that no actual theory can achieve $\Delta_M = 0$ (i.e., zero discretion) for all disputed constitutional provisions. The proof proceeds by showing that each of the three ideal conditions (i)–(iii) is impossible to satisfy fully.

(a) Contested fixation moment. Even if a theory fixes meaning at a historical moment, there is no uniquely determined "moment of fixation" that all interpreters agree upon. Originalists themselves disagree: some argue for ratification (when the Constitution became law), others for the framing (when the text was drafted), and still others for the period of public debate in between. This disagreement is not resolvable by semantic or historical evidence alone; it reflects a normative choice about what counts as the "authoritative" moment. Hence property (i) cannot be satisfied in an uncontested manner.

(b) Semantic underdetermination. Even with a fixed historical moment, the semantic content of the text does not uniquely determine applications to novel fact situations. This is the classic problem of vagueness (Williamson 1994). For any vague predicate (e.g., cruel and unusual), there exist borderline cases where the predicate neither clearly applies nor clearly does not apply. In such cases, the semantic meaning alone cannot yield a unique decision; some extra-semantic judgment is required. Therefore property (ii) (excluding all non-semantic sources) is insufficient to eliminate discretion.

(c) Computational undecidability. Suppose one attempts to resolve ambiguity via a mechanical algorithm $\mathcal{A}$. The problem of determining whether a given interpretation follows from the text and a finite set of canons is, in general, computationally undecidable for sufficiently rich languages. More precisely, the set of valid interpretations of a natural-language legal text under interpretive canons is not recursively enumerable in a formal sense; there is no algorithm that, for every ambiguous provision, halts with a unique output. This follows from the fact that legal language is at least as expressive as first-order logic (which is undecidable), and the canons of construction do not reduce it to a decidable fragment. Hence property (iii) is unattainable.

Because all three ideal conditions are impossible to realize jointly (and in fact each is impossible in practice), no actual theory can achieve zero discretion. The maximum feasible constraint is strictly less than 1, and both originalism and textualism attain the highest achievable constraint ratios among existing theories (as shown in the comparative constraint analysis, Table 5.4). ■

Conclusion of Corollary 2: The ideal of complete constraint is a useful benchmark but is unattainable due to contested fixation, semantic underdetermination, and computational undecidability. Originalism and textualism represent the best approximations available, but they inevitably leave residual discretion.

Proof of Proposition 6 (Meta-Choice Underdetermination)

Proposition 6 (Meta-Choice Underdetermination): Let $\mathcal{M} = \{\text{orig}, \text{text}, \text{struct}, \text{live}, \text{prag}\}$ be the set of interpretive theories. There does not exist a function $G : \mathcal{M} \to \mathbb{R}$ that ranks theories according to their formal properties alone (e.g., consistency, determinacy, parsimony) such that all rational Justices would agree on the optimal theory.

Proof:

I prove by contradiction and by appeal to incommensurability of fundamental values. Assume, contrary to the proposition, that there exists a function G that maps each theory $M \in \mathcal{M}$ to a real number, where $G(M)$ is computed solely from formal properties of the theory – such as internal consistency, degree of determinacy (i.e., how often $\Delta_M = 0$), syntactic parsimony, or mathematical elegance – and that for any two rational Justices, the ranking of theories by G would be identical. That is, the Justice who prefers higher G would choose the same theory regardless of her normative commitments.

Step 1 – Formal properties do not determine a unique ranking

The formal properties of the five theories, as derived in Sections 3–5, include: 1) Consistency: All five theories are internally consistent; no theory entails a contradiction.; 2) Determinacy (constraint ratio): $R_{\text{orig}} \approx 0.68$, $R_{\text{text}} \approx 0.61$, $R_{\text{struct}} \approx 0.43$, $R_{\text{live}} \approx 0.12$, $R_{\text{prag}} \approx 0.09$; 3) Parsimony: Textualism and originalism are more parsimonious (fewer sources of authority) than structuralism, living constitutionalism, and pragmatism; and 4) Mathematical elegance: Not obviously comparable.

Even if we could compute a single number from these properties (e.g., a weighted sum), different Justices might assign different weights to determinacy vs. parsimony. For instance, a Justice who values predictability above all would rank originalism highest (largest R), while a Justice who values adaptability would rank pragmatism highest. The ranking induced by G would depend on the weights, which are themselves normative. Thus no single G computed from "formal properties alone" (without normative weighting) can produce a unique ordering that all rational Justices accept.

More formally, consider two Justices J_A and J_B. Let R_M be the constraint ratio (a formal property). Suppose G is a function of the vector of formal properties $\mathbf{p}_M = (R_M, \text{parsimony}_M, \dots)$. For J_A to agree with J_B, G must be monotonic in each property in a way that respects both Justices' preferences. But J_A might prefer higher R (more constraint), while J_B might prefer lower R (less constraint). Unless G is constant (which would make all theories equally good, contrary to the existence of a ranking), there exist M_1, M_2 such that $G(M_1) > G(M_2)$ but J_B prefers M_2 over M_1. Therefore no single G can command universal rational agreement.

Step 2 – The underlying incommensurability

The deeper reason is that each theory privileges a different ultimate value that cannot be reduced to a common metric without circularity (Raz 1986, 321–66). Specifically: Originalism privileges fidelity to the past (democratic legitimacy through ratification); Living constitutionalism privileges moral responsiveness (justice as evolving moral understanding); Textualism privileges semantic discipline (rule of law as the enacted text); Structuralism privileges institutional coherence (balanced government); and Pragmatism privileges welfare maximization (consequentialist problem-solving).

These values are incommensurable in the sense that there is no cardinal or even ordinal scale that can represent all rational preference orderings over them. For example, a Justice may rationally prefer fidelity to the past over welfare maximization, while another Justice rationally prefers the opposite. Neither preference is internally inconsistent or based on a factual error. Therefore, any ranking of theories that depends on these values cannot be derived from formal properties alone; it must incorporate a normative stance.

Step 3 – Formal impossibility via Arrow-style argument

It is also possbile to prove impossibility using a social-choice analogy. Let the "voters" be the set of possible rational Justices, each with a complete preference ordering over $\mathcal{M}$. A ranking function G that maps the set of formal properties (which are objective and shared by all) to a single real number would, if it existed, induce a complete social ordering over $\mathcal{M}$ independently of individual preferences. That would violate the Pareto principle and independence of irrelevant alternatives unless the formal properties themselves dictate a unique ordering – which they do not, as shown in Step 1.

More concretely, suppose we try to define $G(M) = w_1 R_M + w_2 \cdot \text{parsimony}_M$. The weights w_1, w_2 are not given by formal properties; they must be chosen. Different choices produce different rankings. Since no rational Justice can be compelled to accept one set of weights over another without appealing to values, the required G does not exist. ■

Conclusion

No function of formal properties alone can rank the five interpretive theories in a way that all rational Justices would agree upon. The choice among theories is irreducibly normative, reflecting a prior commitment to incommensurable values. This completes the proof of Proposition 6.

Annex C - Other Theories of Constitutional Interpretation

The five theories formalized above—originalism, living constitutionalism, textualism, structuralism, and pragmatism—represent dominant and well-defined clusters in the contemporary debate. However, the landscape of constitutional interpretation is broader and more variegated. Several additional theories, some overlapping with or refining the core five, merit attention for their influence on judicial practice and academic discourse. This addendum briefly identifies these theories, their core claims, and leading proponents, noting their relationship to the axiomatic framework already developed.

1. Doctrinalism (Modalities of Argument)

Philip Bobbitt's (1982) seminal work identifies six "modalities" of constitutional argument: historical, textual, structural, doctrinal, ethical, and prudential. Doctrinalism, as a stand-alone theory, emphasizes the primacy of precedent and the gradual elaboration of legal rules through case law. It treats the Constitution as a common-law document whose meaning is found in the accumulated body of judicial doctrine, not in text or original understanding alone. This approach is closely aligned with common-law constitutionalism and is often invoked to justify stare decisis. Major defenders include David Strauss (2010), who argues that the living Constitution is best understood as a common-law system, and Frederick Schauer (2009), who analyzes the force of precedent. In our axiomatic terms, doctrinalism would define the constraint set $\mathcal{D}_{\text{doct}}$ as decisions consistent with the best fit to existing precedent, with the objective of maximizing coherence within the doctrinal fabric. It shares living constitutionalism's evolutionary premise but channels change through incremental judicial reasoning.

2. Moral Reading (Dworkinian Interpretivism)

Ronald Dworkin's (1996) "moral reading" is often subsumed under living constitutionalism, but it is sufficiently distinctive to warrant separate mention. Dworkin argues that abstract constitutional clauses (e.g., "equal protection," "cruel and unusual punishment") refer to moral concepts, and Justices must decide cases by constructing the best moral theory that fits and justifies the constitutional text and past decisions. This is not a free-floating moralism but is constrained by the dimensions of "fit" and "justification." The Justice acts as a Herculean interpreter seeking the interpretation that makes the law the best it can be. In the earlier axiomatic model for living constitutionalism, the objective function $\alpha\text{Fit} + (1-\alpha)\text{Morality}$ captures Dworkin's framework, but Dworkin's theory places heavier emphasis on the integrity of the legal system as a whole. Proponents include Dworkin himself and, in a modified form, James E. Fleming (2006).

3. Popular Constitutionalism

Popular constitutionalism challenges the judicial supremacy implicit in most other theories. It holds that the ultimate authority to interpret the Constitution resides not in the courts but in the people and their elected representatives. Constitutional meaning is constructed through political mobilization, social movements, and legislative action; judicial review is either subordinate to popular will or should be exercised with extreme deference to democratic outcomes. Key proponents include Larry Kramer (2004), Mark Tushnet (1999), and, in a more historical vein, Robert Post and Reva Siegel (2006). This theory radically decenters the Justice. In our framework, it would not be a decision procedure for an individual Justice but a meta-constraint: the Justice's admissible decision set $\mathcal{D}_{\text{pop}}$ would be restricted to those interpretations that are endorsed by or at least not in conflict with sustained popular constitutional understandings. The Justice's values V would be largely irrelevant, as the interpretive authority is externalized to the political community.

4. Process Theory (Representation-Reinforcing Review)

John Hart Ely's (1980) "Democracy and Distrust" offers a theory that is partly structuralist but distinct enough to be considered on its own. Ely argues that judicial review is legitimate only when it reinforces the democratic process—by policing the channels of political change and protecting discrete and insular minorities from majoritarian prejudice. The Constitution's open-ended provisions should not be used to impose substantive values but to correct failures in the political market. This theory is a form of structuralism focused on democratic process. In axiomatic terms, the constraint set $\mathcal{D}_{\text{process}}$ would include only those decisions that remedy process defects; substantive value judgments by Justices are impermissible. Ely's theory influenced a generation of scholars and remains a touchstone for debates about judicial restraint and the counter-majoritarian difficulty.

5. Critical Theories (Critical Race Theory, Feminist Jurisprudence etc.)

Critical theories of interpretation, including critical race theory (Bell 1992; Crenshaw et al. 1995) and feminist jurisprudence (MacKinnon 1989; Scales 2006), argue that traditional interpretive methods mask the operation of power and perpetuate systemic inequalities. They contend that constitutional meaning cannot be divorced from the social, racial, and gender hierarchies that shape legal institutions. Interpretation must be attentive to the lived experiences of marginalized groups and must actively dismantle oppression. These theories reject the notion of a neutral interpretive method; they foreground the Justice's situatedness and the political consequences of interpretation. In our framework, they would treat the Justice's values V not as a bias to be minimized but as a necessary lens for revealing injustice. The objective function would explicitly prioritize the advancement of equality and the dismantling of subordination, with few, if any, formal constraints beyond the imperative to do justice. Key references include Derrick Bell, Kimberlé Crenshaw, Catharine MacKinnon, and Robin West.

6. Natural Law and Moral Traditionalism

A tradition with roots in the founding era and natural-rights philosophy holds that the Constitution incorporates certain pre-political moral truths or principles of natural law, which Justices must enforce even if not explicitly stated in the text. This view was influential in early American jurisprudence (e.g., Calder v. Bull, 1798) and has been revived by some natural-law scholars (George 1999). It differs from originalism in that the fixed meaning is not merely semantic but moral; it differs from Dworkin's moral reading in that the moral content is discovered, not constructed. In our axiomatic model, it would impose a constraint that d must be consistent with the overarching moral order, a constraint that is objective in theory but highly contested in application. Judicial discretion is ostensibly limited by the moral reality the Justice is bound to discern, but in practice the identification of that reality is deeply value-laden.

7. Comparative Constitutional Interpretation

While not a full interpretive theory in the domestic sense, the increasing use of foreign and international law in constitutional adjudication has generated a methodological stance that some treat as a distinct approach. Proponents (e.g., Breyer 2005; Jackson 2010) argue that interpreting the Constitution in light of global human rights norms and the experiences of other democracies enriches understanding and promotes a more cosmopolitan jurisprudence. Opponents (Scalia, originalists) view this as illegitimate. In an axiomatic model, this approach would expand the context vector C to include foreign and international legal materials, and might adjust the objective function to favor interpretations that align with transnational consensus. It is a modification of living constitutionalism with a global orientation.

Concluding Remarks on the Taxonomy

The five theories modeled in the main paper are not exhaustive. Doctrinalism, moral reading, popular constitutionalism, process theory, critical theories, natural law, and comparative approaches each illuminate different dimensions of constitutional meaning and judicial role. Many of these theories can be seen as refinements, radicalizations, or alternative specifications of the core five. For instance, process theory is a species of structuralism; doctrinalism is a variant of living constitutionalism; moral reading is a philosophical version of living constitutionalism; popular constitutionalism challenges the Justice-centered premise of all models. The axiomatic framework developed earlier is flexible enough to accommodate these additional theories by adjusting the objective function and constraint set accordingly. The fundamental insight remains: no theory can eliminate the interplay of textual ambiguity and subjective values; each merely structures that interplay in a different way, reflecting a distinct vision of constitutional legitimacy.

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